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# Technical Report No. 26

# Sterilizing Filtration of Liquids

Revised 2025

PDA Parenteral Drug Association

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# Technical Report No. 26

# Sterilizing Filtration of Liquids

Revised 2025

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# 1.0 Introduction

Sterilizing filtration is the process of removing microorganisms* from a fluid stream without adversely affecting product quality ***(1-4)***. This technical report is intended to provide a systematic approach to selecting and validating the most appropriate filter for liquid sterilizing filtration applications.

PDA's original *Technical Report No. 26: Sterilizing Filtration of Liquids*, (TR 26) published in 1998 described the use and validation of sterilizing filtration to a generation of pharmaceutical scientists and engineers. A revision of the original report was developed in 2008 in response to enhancements in filtration technologies and recent additional regulatory requirements within the pharmaceutical industry at that time. This current revision has been implemented to reflect current best practices and regulatory expectations, as well as clarify the roles of filter suppliers and end-users. References are provided for scientific publications and international regulatory documents where more detail and supportive data may be found.

When membrane filters entered the commercial scale, 0.45 µm-rated membranes were considered "sterilizing-grade" filters and were used successfully in the sterilizing filtration of parenterals. These filters were qualified using *Serratia marcescens* as a standard bacterium for qualifying membranes used for water quality testing. In a paper published in 1960, however, Dr. Frances Bowman of the U.S. Food and Drug Administration (FDA) observed a 0.45 µm "sterile-filtered" culture medium to be contaminated with an organism subsequently shown to repeatedly penetrate 0.45 µm-rated membranes in small numbers at challenge levels above $10^4 – 10^6$ colony forming units (CFU) per $cm^2$ ***(5)***. This led to the development of American Society for Testing and Materials (ASTM) F838, a standard test method for evaluating sterilizing-grade membrane filters ***(6)***. Challenge organisms are discussed further in **Section 6.0**.

***Note:** Sterilizing filtration is not intended to remove viruses.

## 1.1 Purpose & Scope

The primary objective of the TR 26 revision team was to develop an updated scientific technical report on sterilizing filtration. This report does not always address region-specific regulatory expectations but it provides up-to-date scientific recommendations for use by industry and regulators in establishing a sterilizing filtration policy. This report should be considered a guide and is not intended to establish mandatory standards for sterilizing filtration. Concepts presented in TR 26 pertain to processes in which filtration is used to render a liquid sterile, with all accompanying assurances, and may not be universally applicable to all filtration processes, such as upstream bioburden reduction steps.

The revision team was composed of European and North American industry professionals to provide a diverse perspective, thus ensuring that the methods, terminology and practices of sterilizing filtration presented are reflective of sound science and can be used globally. This report underwent review by global technical peers that included feedback from experts from the Americas, the Asia-Pacific region and Europe.

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# 2.0 Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used vary, for instance, among companies and geographic location, the definitions described below are consistently used within this technical report. Where a definition is based on another published source, the source is cited.

***Adsorption***
The retention of solutes, suspended colloidal particles or microorganisms to fluid contact surfaces, e.g., the surfaces of pores in filtration membranes.

***Aseptic***
Conditions that minimize viable microbial population.

***Assay***
Analytical method used to determine purity or concentration of a specific substance in a mixture.

***Bacterial Retention Study***
An experimental test study used to demonstrate that a filter can effectively retain a defined level of challenge microorganism under specified conditions.

**Note:** This is also referred to as "Bacterial Challenge Test".

***Bioburden***
The type and number of microorganisms per unit of material ***(7)***.

***Bracketing Approach***
A validation method that tests the specification extremes of a process or product. The method assumes the extremes will be representative of all the samples between the extremes.

***Bubble Point***
The measured differential gas pressure at which a wetting liquid (e.g., water, alcohol or product) is pushed out of the largest pores of a wetted porous membrane and a steady stream of gas bubbles or bulk gas flow is detected.

**Note:** This is also referred to as "Transition Point".

***Bubble Point Test***
A test to indicate the maximum pore size of a filter. The differential gas pressure at which a liquid (usually water) is pushed out of the largest pores and a steady stream of gas bubbles is detected from a previously wetted filter under specific test conditions. Used to test filter integrity with specific, validated, pressure values, wetting liquids and temperatures for specific pore-size (and type of) filters.

***Capsule Filter***
A self-contained filter device or assembly.

***Cartridge Filter***
A filter device requiring a housing for use.

***Compatibility***
Proof that no adverse interaction between the filter, process fluid and filtration conditions has occurred.

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***Critical Process Parameter (CPP)***
A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces the desired quality ***(8)***.

***Diffusive Flow***
The movement of a dissolved gas across a liquid-wetted membrane based on concentration (e.g., gas pressure) differential.

***Diffusive Flow Test***
A test to determine the integrity of a filter based on a measurement of the diffusive flow rate of a gas through wetted filter membrane at a set pressure limit.

**Note:** "Diffusive Flow" and "Forward Flow" are generally synonymous terms, "diffusive flow" will be used in this document.

***Double Filtration***
Filtration through two filters configured in series whereby both filters are required to achieve a sterile effluent as determined by the product and process-specific bacterial retention validation study (i.e., both filters must meet post-use integrity test specifications).

**Note:** This is also referred to as "Additive Filtration".

***Downstream Side (of a Filter)***
The filtrate or outlet side of the filter.

***Effective Filtration Area (EFA)***
The total surface area of the filter available to the process fluid.

***Effluent***
Fluid that flows out of a process step.

***Endotoxin***
The major constituent of the outer membranes of Gram-negative bacteria composed of lipid A, the core polysaccharide, and the O-antigen polysaccharide. Endotoxin is also known as lipopolysaccharide (LPS).

***Extractable***
A chemical component that is removed from a material by application of an artificial or exaggerated force (e.g., solvent, temperature, time).

***Filter (noun)***
A device used to remove particles from a fluid process stream that consists of a porous medium and a support structure.

Porous material through which a liquid or gas is passed to remove viable and nonviable particles ***(6)***.

***Filter (verb)***
To pass a fluid through a porous medium whereby viable and nonviable particles are removed from the fluid.

***Filterability Test***
A test to determine the suitability and sizing of a filter with a given fluid that typically involves measuring the rate at which a liquid can be filtered through a specific filter and the maximum filter capacity before the filter becomes clogged/fouled.

***Filter Efficiency***
A measurement of how well a filter retains particles. It is usually expressed as the percentage, or ratio, of the retention of particles of a specific size by a filter.

***Filter Element***
The basic filter unit from which cartridges or capsules are assembled.

***Filtrate***
Fluid that has passed through a filter.

***Filtration***
The process by which particles are removed from a fluid by passing the fluid through a porous material.

***Flow Rate***
The volumetric rate of flow of a solution, expressed in units of volume per time (e.g., L/min or gal/day).

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***Flux***
The rate of filtrate flow divided by the membrane area.

***Forward Flow Test***
See "Diffusive Flow Test".

***Fouling***
The result of solutes blinding or blocking membrane pores. It is observed as a decrease in the flux (at constant pressure) or an increase in the filtration differential pressure (at constant flux).

***Gamma Irradiation***
Ionizing radiation, generated by decay of a radioactive nucleus, that can be used to sterilize a material.

***Gauge Pressure***
Gauge pressure is the difference between a given fluid pressure and that of the atmosphere.

***Hydrophilic***
Literally "water loving", a filter that will wet with aqueous solutions to allow flow at low pressure differential.

***Hydrophobic***
Literally "water fearing", a filter that repels aqueous and other high surface tension fluids and therefore cannot be wetted unless subjected to a high-pressure differential. When prewetted with low surface tension fluid, such as alcohol, the membrane will allow water flow at low pressure differential.

***Influent***
Fluid that flows into a process step.

**Note:** This is also referred to as "Feed".

***Integrity Test***
A nondestructive physical test that can be correlated to the bacterial retention capability of a filter/filter assembly ***(6)***.

***Leachable***
A chemical component that migrates from a contact surface into a drug product or process fluid during storage or normal use conditions.

***Materials of Construction (MoC)***
Polymers or other materials that make up the components of the filter.

***Medium***
In filtration, the porous material that retains particles as a fluid passes through during the process of filtration.

***Membrane***
A thin, microporous medium used to remove particles and microorganisms from a fluid stream under pressure.

***Module***
Filter element that is incorporated into a cartridge or capsule.

***Multi-Round Housing***
Filter housing designed to allow multiple filter elements installed in parallel to increase the effective filtration area (EFA).

***Non-Fiber Releasing***
Refers to a filter that does not shed fibers into the filtrate.

***Particle***
Any discrete unit of material structure; a discernible mass having an observable length, width, thickness, size and shape.

***Particulate***
Relating to, or occurring in the form of, particles.

***Pore***
The channel(s)/path(s) in a membrane through which a fluid may pass.

***Porosity***
The ratio of void volume to bulk volume of the filter media.

***Prefilter***
Any filter placed upstream of the sterilizing filter(s).

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***Pressure***
Force applied per unit area usually expressed as psi, mbar, kPa or $kg/cm^2$.

***Back Pressure***
Pressure applied downstream of a filter or other piece of equipment, via restriction in flow.

***Differential Pressure***
The difference in pressure between the upstream (feed or influent) and downstream (effluent) sides of the filter. (May be modified with the following terms: applied differential pressure, available differential pressure, clean differential pressure, dirty differential pressure, initial differential pressure, or maximum differential pressure.)

**Note:** This is also referred to as "Delta P (Δ P)", "Psid", or "Pressure drop".

***Inlet Pressure***
The applied pressure entering the upstream side of the filter.

**Note:** This is also referred to as "Influent", "Upstream Pressure" or "Line Pressure".

***Outlet Pressure***
The pressure exiting the downstream side of the filter.

**Note:** This is also referred to as "Effluent Pressure" or "Downstream Pressure".

***Redundant Filtration***
A type of serial filtration where a second sterilizing filter is used as a backup in the event of an integrity failure of the primary sterilizing filter.

***Serial Filtration***
Filtration through two or more filters of the same or decreasing pore size, one after the other.

***Sterilization***
Validated process used to render a product free of viable microorganisms.

**Note:** In a sterilization process, microbiological death or reduction is described by an exponential function. Therefore, the number of microorganisms that survive a sterilization process can be expressed in terms of probability. While the probability may be reduced to a very low number, it can never be reduced to zero.

***Surface Tension***
The tendency of the surface of a liquid to contract to the smallest area possible under defined conditions. It is expressed as dynes per centimeter**.**

***Surfactant***
A soluble compound that reduces the surface tension of a liquid or reduces interfacial tension between two liquids (causing formation of micelles), or between a liquid and a solid.

***Throughput***
The amount of solution that passes through a filter. It is described as volume through the membrane area.

**Note:** This is also referred to as "Capacity".

***Upstream***
The influent, or inlet side of the filter.

***Validation***
A documented program that provides a high degree of assurance that a specific process, method or system will consistently produce a result meeting predetermined acceptance criteria.

***X-ray Irradiation***
Ionizing radiation, generated by an electron beam accelerator coupled with a metal target, that can be used to sterilize a material.

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## 2.1 Abbreviations

| | |
|---|---|
| **CCS** | Contamination Control Strategy |
| **CFU** | Colony Forming Unit |
| **EFA** | Effective Filtration Area |
| **MoC** | Materials of Construction |
| **NOR** | Normal Operating Range |
| **PD** | Process Development |
| **PUPSIT** | Pre-Use, Post-Sterilization Integrity Test |
| **SIP** | Sterilize in Place or Steam in Place |
| **SLB** | Saline Lactose Broth |
| **SUS** | Single Use System |
| **TOC** | Total Organic Carbon |

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# 3.0 How Filters Work

Liquid filters are widely believed to function by permitting fluid passage through their pores, while retaining particles, such as microorganisms, that are too large to fit through these apertures. At first, one might assume this is accomplished entirely by size-exclusion of the particles on the upstream side of a screen or sieve-like filter membrane. However, this oversimplification ignores many complexities of the actual mechanisms of particle retention. Although microporous membranes are very thin to the naked eye, the thickness of the membrane represents a long path from the microscopic perspective of bacteria or other particles in the fluid stream passing through the membrane. Sterilizing-grade filtration membranes do not typically consist of cylindrical pores of well-defined diameter. Rather, the membrane is made up of a sponge-like depth of filter material, and the porous pathways between them, across a distribution of sizes. Large particles completely outside this pore size distribution are certainly excluded at the upstream surface of the membrane (i.e., surface screening). But for more marginally sized particles, size-exclusion also includes entrapment within the depth of the filter matrix.

Additional retention mechanisms beyond simple size exclusion also apply within the depth of the membrane: adsorptive sequestration occurs when particles small enough to pass through filter pores are still captured by the filter, through various chemical or physical effects that cause particles to bind to the filter material that retains them. The effectiveness of adsorptive-sequestration mechanisms can depend on the surface chemistry of the filter and the particle (e.g., a microorganism) under the applied filtration conditions. Many different operational conditions govern a filter's adsorptive removal of particles including differential pressure, flow rate, number of particles present, binding-site saturation, and the liquid vehicle's makeup in terms of its surface tension, pH, and ionic strength, among other factors.

If each particle challenging the filter is too large to pass through any of the pores, the number of particles does not matter – all will be retained by surface-screening. For such large particles, filter efficiency (the completeness with which a filter retains particles confronting it) is independent of many of the complex factors listed above, as long as the particle and pores are not deformed. But for small microorganisms, which can grow, shrink, deform, and divide, the additional retention mechanisms and process factors become important. For sterilizing applications, a user must recognize that complete retention of microorganisms in a feed stream may become a probabilistic feature of the process and filter, depending on the factors discussed above. For this reason, the details of each specific sterilizing-filter application are important and must be validated to claim the use of filtration as a mode of sterilization. Fortunately, modern sterilizing filter design and manufacturing processes incorporate robust controls and significant safety factors that reliably assure sterile filtrates in many applications.

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## 3.1 Pore Size Rating and Filter Nomenclature

Because of the complex sponge-like distribution of pore sizes in membrane filters, characterization of a filter by a simple pore-size (diameter) rating has been controversial. Various methods have been proposed and used by filter manufacturers for filters of varying porosity. For sterilizing filters specifically, a consensus has arisen in the industry to define sterilizing-grade filters not by physical pore size measurement, but by its ability to reproducibly retain $10^7$ CFU per $cm^2$ *Brevundimonas diminuta* according to the ASTM F838 standard ***(6)***. Due to historical reasons, it is the successful demonstration of this retention challenge that may grant a filter (0.2 µm or smaller) a sterilizing-grade rating ***(9)***. To reiterate, quantified pore-size ratings, such as 0.2 µm, may be assigned without any physical measurement or characterization of pore-size but based simply on the ability of the filter to pass the standard bacterial retention study. This characterization has often been used to market filters as sterilizing-grade. As described in **Section 6.0**, actual use for critical liquid sterilization applications (where a sterile claim is made for the filtrate) also requires process-specific validation, only then should a filter be described as a "sterilizing filter."

Due to the important nuances of these terms, recommended uses of these terms are summarized in **Table 3.1-1**.

**Table 3.1-1 Recommended Terminology Related to Sterilizing-Grade Filters**

| Term | Definition |
|---|---|
| 0.2 µm or 0.22 µm Rated Filter | These pore-size ratings may be assigned by filter suppliers based on successful completion of the ASTM F838 retention test (synonymous with "sterilizing-grade filter"). This rating may also be assigned based on other criteria and may be assigned to filters intended only for bioburden reduction or filters that have not passed the ASTM F838 retention test. For this reason, "sterilizing-grade filter" is a more precise term for describing filters designed for sterilization of liquids.<br><br>**Note:** 0.22 µm and 0.2 µm are considered interchangeable nominal pore-size ratings ***(7)***. |
| Sterilizing-Grade Filter | Filters, typically with an 0.22 µm /0.2 µm pore size rating or smaller have been shown by the manufacturer to have successfully completed the standard bacterial retention study, that is, complete retention of $10^7$ CFU per $cm^2$ *B. diminuta* under standard conditions described by ASTM F838.<br><br>**Note:** This is a claim about a filter (as marketed by the supplier) independent of any particular application. |
| Sterilizing Filter | This is an additional claim about the function of a filter to actually sterilize a fluid in a specific application, and it depends on successful execution of process-specific validation and other controls in addition to the standard ASTM F838 challenge.<br><br>**Note:** When using this term, a sterile claim for the filtrate is being made. |

**Note:** The term "sterile filter" is often used to refer to a "sterilizing filter"; however, that should generally be avoided due to ambiguity whether it refers to the purpose of the filter (sterilizing a fluid) or the state of the filter (the filter itself is sterilized, e.g., via steam or irradiation).

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**Figure 3.1-1** attempts to further clarify these terms with a Venn diagram. This diagram should be taken as a generalization only, as exceptions exist.

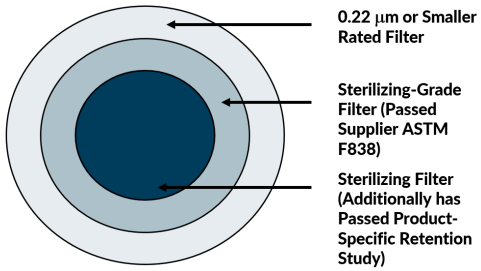


**Figure 3.1-1 Sterilizing Filter Terminology: Venn Diagram**

## 3.2 Evolution of the Sterilizing Filter

Membrane filtration has been used for decades and became a prevalent mode of sterilization especially with the introduction of heat-sensitive compounds that typically cannot be terminally sterilized.

Membrane filters are used for microbial control in multiple different applications and manufacturing process steps, such as the filtration of cell-culture media, and intermediate steps like purification column protection, buffer filtration, and bioburden reduction as well as final sterilization.

These different filtration steps typically have different requirements for an optimal filtration process, which means that modern membrane filters are optimized for specific applications, with an emphasis on robust and reliable retention of organisms. For the most critical application of final product sterilization, modern sterilizing-grade membrane filters are designed and built to meet the strict requirements of the biopharmaceutical and pharmaceutical industries and regulatory authorities ***(9-13)***. These filters have undergone continuous optimization and development, resulting in a robust filter designs with well-controlled attributes such as ***(14)***:

- **Steam Sterilization Resistance:** Filters can withstand high temperatures making them suitable for sterilization by in-line steam or autoclaving, even for multiple cycles. The temperature/differential pressure profiles of a filter are well-established, which supports the appropriate operating conditions during steam sterilization, preventing any damage.
- **Irradiation Resistance:** Filter capsules may also be supplied presterilized (e.g., if part of a single-use system (SUS) assembly/manifold). These filters are compatible with irradiation sterilization methods, such as gamma or x-ray irradiation. Presterilized filters should be validated to show proof that they are sterilized, even when they are sterilized within the assembly, and the SUS is proven to be compatible with the sterilization method ***(15)***.
- **Minimal Extractable Substances:** Even when used straight out of the box, these filters release extremely low levels of extractables that have been published and are detailed in qualification documentation from the filter manufacturers.
- **Mechanical Resistance:** Modern sterilizing filters can withstand high-differential-pressure conditions, as validated by the filter manufacturers ensuring their durability and reliability.
- **Low Unspecific Adsorption**: Most filters have been optimized to greatly reduce the amount of unspecific adsorption, which is important to avoiding the retention of target proteins or other important fluid components, for example polysorbate.
- **Optimal Total Throughputs:** Some filters are designed to maximize volume of filtration, effectively reducing the cost per liter of filtered material. Highly asymmetric and heterogenous double-layer filter designs are throughput-focused filter types.

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- **Flow Rates:** Filters are engineered to achieve efficient flow rates, which can reduce processing times or the required filtration area. Engineering features such as high pleat density, large effective filtration area (EFA), low-pressure drops, and single-layer membranes are available for flow rate optimization.
- **Robust Integrity-Testing:** Sterilizing-grade filters are required to be tested for their integrity using nondestructive methods such as diffusive flow, pressure decay, or Bubble Point tests. These tests have been successfully used for decades and are a reliable, essential filter integrity test.
- **Full-Qualification Documentation:** Filters are accompanied by comprehensive qualification documentation, which lists a multitude of tests (e.g., endotoxin release, biological reactivity) that ensure they meet the necessary regulatory standards.
- **Individual Process Validation:** The filters undergo process validation specific to the drug product being filtered under process conditions, confirming their compatibility, performance and retentivity.
- **100% Integrity-Tested:** Every sterilizing-grade filter is individually integrity-tested by the filter manufacturer as a release criterion for the product.
- **Transportation and Shelf Life:** Filter suppliers also recommend necessary shelf-life data for filtration and single-use assembly products and validate the packaging and transportation of these components to ensure the products are not damaged.
- **Membrane Casting Batch Consistency:** Filter supplier implemented controls and automation to assure that the membrane casting process consistently delivers the same membrane quality and performance.

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# 4.0 Filter Qualification, Design, and Characterization

Filter manufacturers typically provide the information discussed in this section, which should be considered by the end user for their risk assessment when selecting a sterilizing-grade filter. Since bacterial retention studies and filter integrity testing are important components of manufacturer-supplied information, those topics are covered separately in **Section 6.0** and **Section 7.0**, respectively.

Sterilizing-grade filters are available in a variety of sizes, configurations (e.g., flat sheets, capsules, cartridges) and membrane chemistries so that the filter user can select the most appropriate filter for a particular application. Commonly used membrane chemistries include polyvinylidene fluoride (PVDF), polysulfone (PS), polyethersulfone (PES), nylon, cellulose esters, polytetrafluoroethylene (PTFE), polyester and polypropylene. Not only do different chemistries impart differences in flow characteristics and filtration performance, but they also affect the levels of filter extractables and leachables, the thermal and physical properties of the filter, and the interactions with the process stream (as determined via compatibility and adsorption-testing). Once the membrane is manufactured into its final format, it is evaluated by the filter manufacturer with respect to flow rate and pressure drop, operating limits of temperature and pressure, extractables, compatibility, cleanliness, sterilization conditions, integrity, and bacterial retention applying defined test conditions.

Since sterilizing-grade filters are a critical part of any aseptic process, a significant amount of testing and documentation is conducted by filter manufacturers to qualify their filters. Filter support documentation may include a validation or qualification guide, FDA Drug Master File (DMF), product literature, specification sheets, technical bulletins and application notes. Pharmaceutical-grade filters are also often accompanied by an individual certificate listing the filter part, lot numbers and release criteria. The supporting documentation from filter manufacturers can help the end users in their assessment for qualifying filters and supporting regulatory inquiries.

## 4.1 Filter Qualification and Validation

Filter manufacturers typically publish results of tests performed according to applicable compendial methods to qualify the filter as suitable for pharmaceutical applications. This qualification documentation supports, but does not replace, performance qualification as a part of process validation conducted by the filter user. **Table 4.1-1** lists qualification and lot-release tests on membranes and devices commonly performed by manufacturers and validated by filter users.

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**Table 4.1-1 Qualification and Validation Recommendations**

| Criteria | Filter User | Filter Manufacturer | |
|---|---|---|---|
| | | Membrane | Device |
| Bacteria Retention in Water or Saline Lactose Broth (SLB) with Integrity Test Correlation in Water or Solvent | - | Q, L | Q, L |
| Bacteria Retention in Product with Respective Process | V | - | - |
| Chemical Compatibility, Effects on Filter Integrity | V | Q | Q |
| Extractables | E | Q | Q |
| Leachables | E | - | - |
| Sterilization Method, Effects on Filter Integrity | E | Q | Q |
| Integrity Test (Water or Solvent) | E | Q, L | Q, L |
| Integrity Test (Product) | V | - | - |
| Toxicity Testing – Bioreactivity Tests | - | Q | Q |
| USP Bacterial Endotoxin Test | E | - | Q, L |
| USP Particulate Matter | E | - | Q |
| USP Nonfiber Release | E | - | Q |
| Flush Volume | E | - | Q* |
| Filter Selection/Fit for Purpose | E | - | - |
| Impact to Product Critical Quality Attributes (CQAs) | E | - | - |

L = Filter Manufacturer's Lot Release Criteria

Q = Filter Manufacturer's Qualification

V = Process Validation Testing (Rationale can be provided for scale-down testing)

E = Evaluate the Need for Additional Testing, Document Technical Assessment

+ = Fully Assembled Filter with Membrane, Filter Components and Plastic Housing (if applicable)

* = Typically Using Water

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Further details of the tests listed in **Table 4.1-1** are provided below:

- Bacterial retention of a filter is initially qualified by the filter manufacturer in saline lactose broth (SLB), water or another suitable carrier fluid and then again by the filter user under process conditions for critical sterilizing-filtration applications. In each case, 100% retention is correlated to an integrity test using a standard reference fluid (e.g., water or alcohol). During the membrane development process, filter manufacturers define the correlation between integrity testing and bacterial retention following ASTM F838 methodology ***(6)***.
- The chemical compatibility between the filter and product can be qualified by the filter manufacturer. Filter-user testing is often necessary to confirm compatibility of the process fluids with the filter and operation conditions. Tests such as product-based integrity and bacterial retention may be used to assess chemical compatibility.
- Extractables may be qualified by the filter manufacturer using model solvents and specific laboratory conditions and may also be determined by the filter user for product-specific fluids under simulated processing conditions (e.g., temperature, duration). Based on the results obtained from the extractables study, the filter user should evaluate the need for testing of leachables. The leachables study should build upon the identified extractables and be tested with the product in scope or a representative surrogate.
- Filter safety testing (i.e., biological reactivity) is conducted by the filter manufacturer and does not need to be repeated by the filter user ***(16)***.
- The capability of a filter to be sterilized and the maximum sterilization conditions recommended for the filter are qualified by the filter manufacturer. The filter user is responsible for validating the sterilization method used in the process*
- Particulate matter, nonfiber release, oxidizable substances, total organic carbon (TOC), conductivity testing and flow rates may be qualified by the filter manufacturer. The filter user should determine if additional testing is required due to differences in processing conditions.
- Filter selection done by the end user will look at filter sizing, filter type based on process scale and design, and impact on product critical quality attributes (CQA) by evaluating compatibility (e.g., adsorption, reaction, degradation) through contact with the filter unit.
- Since solvents could be used as wetting agents to determine if the integrity test failed due to wetting problems, it would be advisable to determine the appropriate amount of flush volumes needed.

***Note:** Sterilization of filter capsules by gamma or x-ray irradiation is generally validated by the filter manufacturer.

## 4.2 Filter Design

Sterilizing-grade membranes are employed in a wide variety of sizes and filter formats, including cartridges, capsules and disc styles to suit many scales and types of applications. Membranes can also be pleated to increase the surface area, and this can be placed in either a cartridge or a capsule filter. **Figure 4.2-1** shows examples of filter cartridges, capsules and components.

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A B C

**Figure 4.2-1 Filter Cartridge (A), Filter Capsule (B), Filter Components (C)**

The design of filter elements is important because design features like membrane pleat shape and density determine the EFA which, in turn, influences the flow rate and total throughput (i.e., the larger the EFA, the higher the flow and/or throughput). If the pleat density is too high, the free flow path is restricted, and the flow is reduced; if the pleat density is too low, the EFA might be too low and the flow restricted. There may be a difference in the pleating geometry used in a scale-up study in process development (PD) and the pleating geometry of the process-scale device, since high-EFA pleating geometries are typically used in process-scale devices only. Potential differences in performance should be evaluated during PD (see **Section 5.4.1** for additional information).

Depending on the filtration needs (e.g., product, process, equipment or facility constraints), different setups and filter types can be employed. **Figure 4.2-2** shows different filter types and configurations.

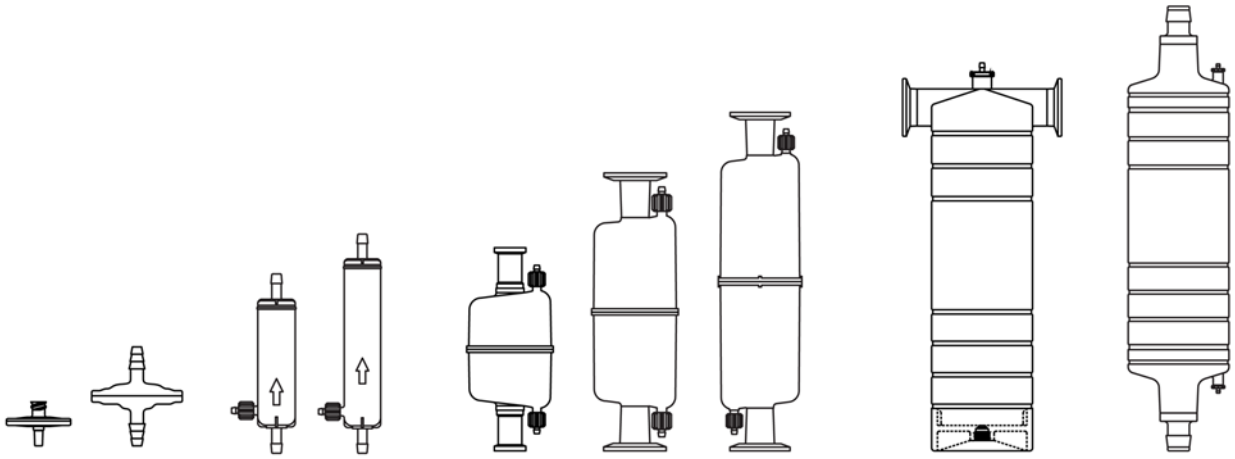

**Figure 4.2-2 Different Filter Types and Configurations**

**Figure 4.2-3** and **Figure 4.2-4** are cutaways that reveal the internal components of a typical cartridge and capsule. Filter capsules, composed of filter elements preinstalled and sealed in plastic housings, are compact and relatively lightweight. Depending on the orientation and configuration of the capsule or filter-housing, the fluid may enter from the bottom or from the top of the capsule or housing and flow around the outside of the internal filter module, pass through the pleated filter membrane to the interior core, and exit the outlet.

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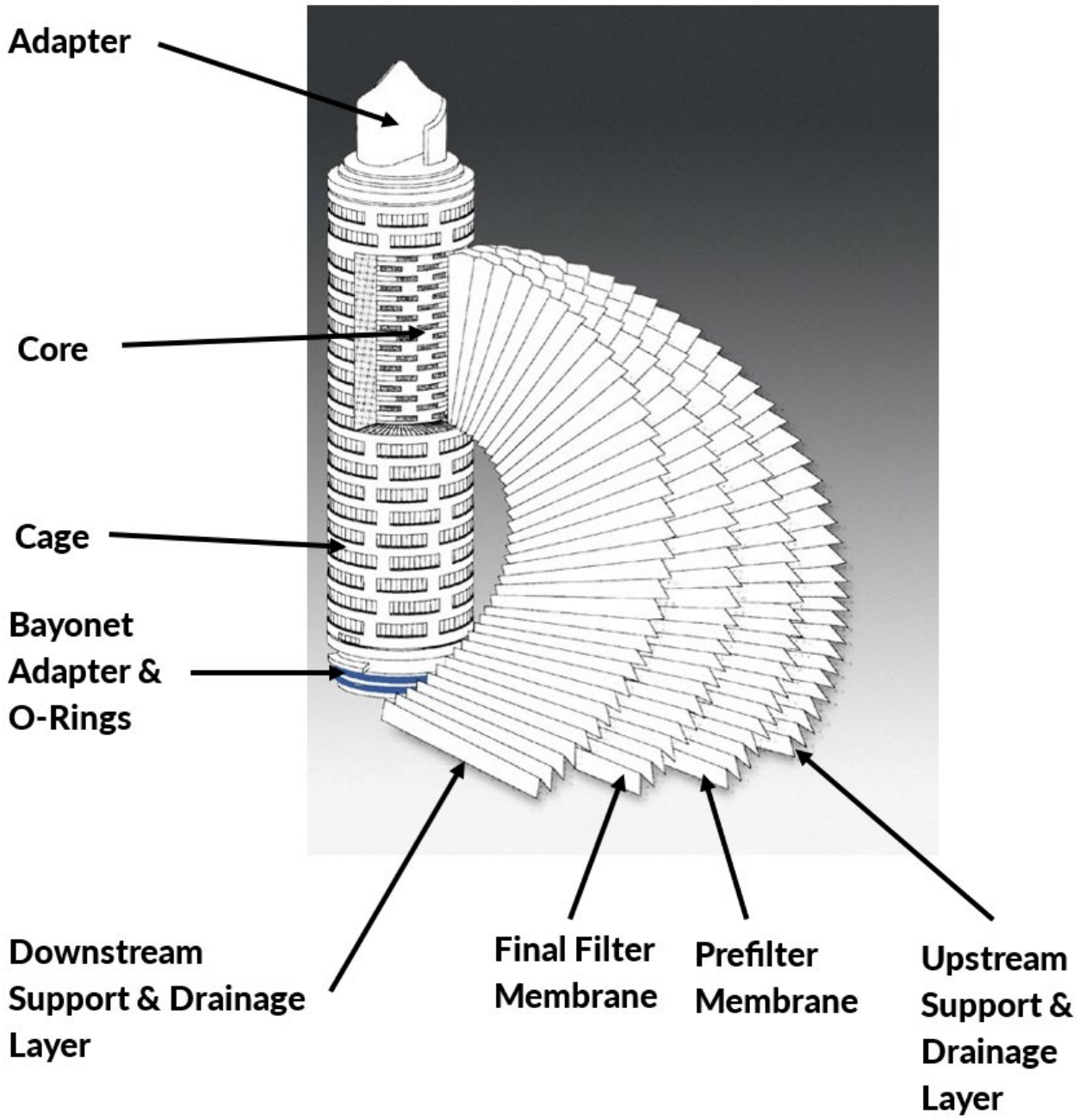


**Figure 4.2-3 Cartridge Internal Components**

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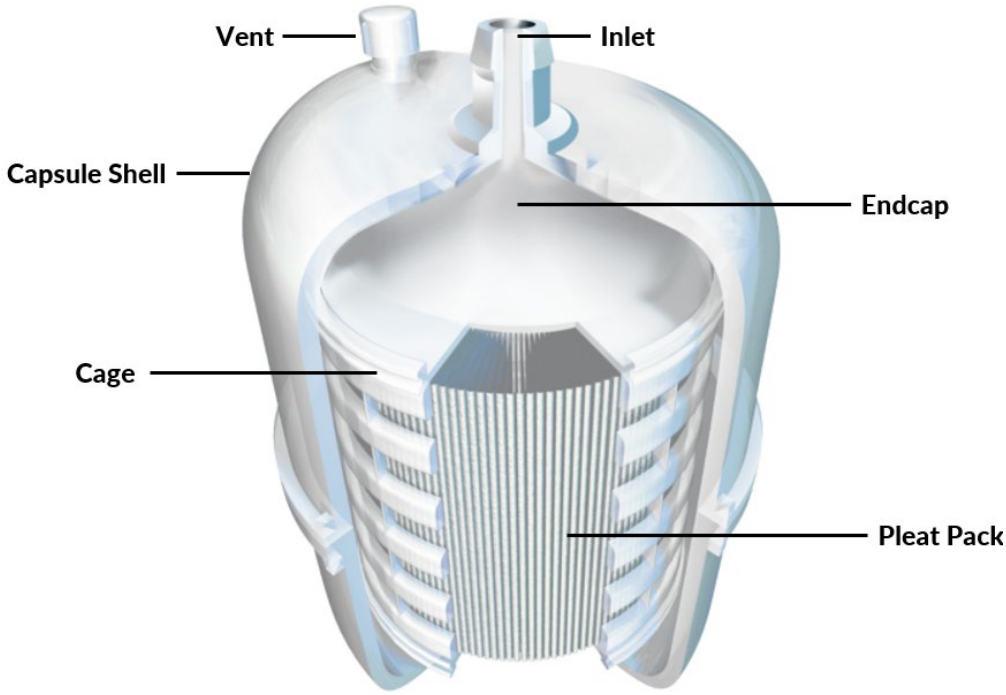


**Figure 4.2-4 Capsule Internal Components**

Stainless-steel filter housings and disposable filter capsules are available in two configurations: T-style or in-line style (**Figure 4.2-5A** and **Figure 4.2-6A,** respectively). The T-style housings have the inlet and outlet pipe adjacent to each other at the bottom plate of the housing (**Figure 4.2-5B**). The inlet pipe in the in-line-style housing is positioned at the dome of the housing and the outlet pipe on the bottom plate (**Figure 4.2-6B**). Most housings and capsules also have vent ports for removing air.

**Figure 4.2-5A Filter T-Style Housing**

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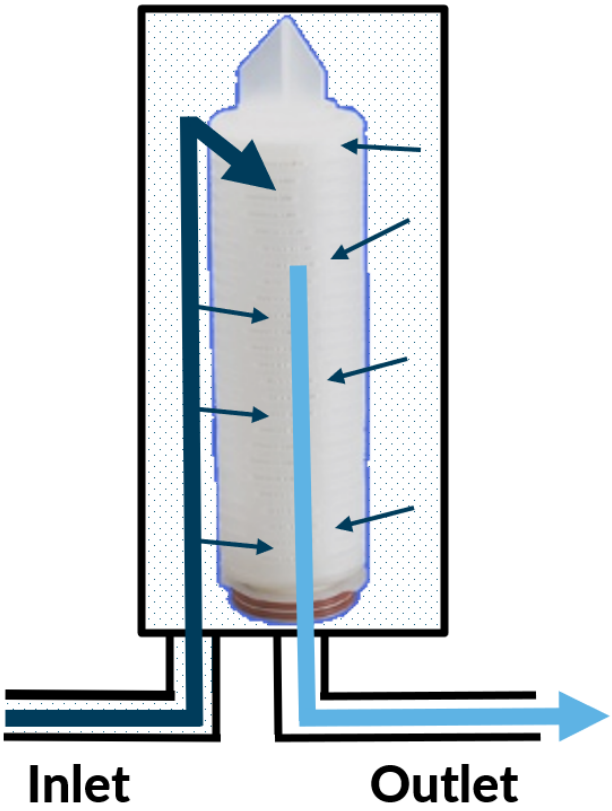


**Figure 4.2-5B Flow Diagram for T-Style Filter**

**Figure 4.2-6A Cutaway View of an In-line Style Stainless-Steel Filter Housing with Cartridge**

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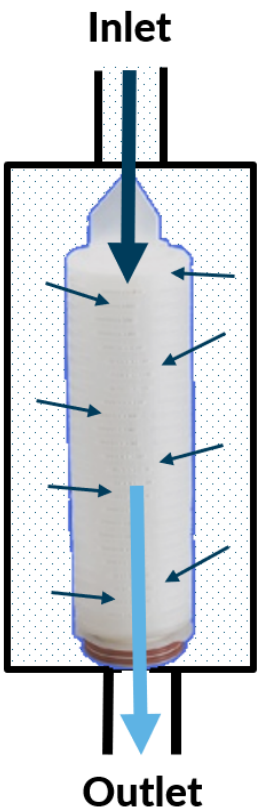


**Figure 4.2-6B Flow Diagram for In-Line Filter**

In addition to housing styles, the filter elements themselves have multiple adapter options (**Figure 4.2-7**) for fitting the filter into the housing. The most common filter-cartridge adapter is a double O-ring, double-bayonet adapter. This adapter type is twisted into place on the base plate to ensure a tight seal between the filter element and the housing base. As housing-base tolerances may vary, it is imperative that the specific filter and housing provide the proper seal throughout the filtration operation (e.g., from steam-in-place (SIP) through post-use integrity-testing).

**Figure 4.2-7 Cartridge Adaptor Options**

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## 4.3 Filters in Single-Use Disposable Systems

It is common to build transfer sets of filters integrated with tubing, sensors, and connectors that are presterilized (e.g., gamma, x-ray) typically used together with other single-use technology or in hybrid systems together with stainless steel equipment. Furthermore, the assembling of multiple filters in ready-to-use assemblies has also become common. These systems offer some advantages over conventional stainless-steel equipment including the:

- Ability to obtain presterilized
- Reduction in setup time
- Elimination of cleaning
- Elimination of cleaning validation
- Elimination of risk of cross-contamination
- Elimination of filter housing maintenance
- Reduction in operator exposure due to closed systems
- Reduction of drug product contamination from the external environment
- Reduction of the operator's exposure to the drug substances

Implementation of SUS should follow a risk-based approach that considers product and process compatibility and demonstrates robustness, suitability, and consistent performance under intended use conditions. (Refer to PDA's *Technical Report No. 66: Application of Single-Use Systems in Pharmaceutical Manufacturing* for additional information.)

Depending on the processing considerations and setup requirements, the filter user chooses the best configuration (capsule, cartridge in stainless-steel housing, or hybrid) for the filtration operation. Use of disposable capsules obviates the need for cleaning validation and equipment amortization, which may reduce operating costs. Another advantage is that operator contact with potentially hazardous products is minimized. The use of stainless steel must be balanced against the use of capsule filters, which can be single use and can reduce utilities and validation efforts. Different optimization can be performed based on environment (waste), energy use, or financial analysis.

## 4.4 Filter Toxicity and Biocompatibility

One important aspect of selecting a filter is assessing the safety of the filter medium and/or device. Filter manufacturers should provide information about the origin and toxicity of the filter components, including their origin of animal-derived components.

### 4.4.1 Toxicity

Filters must not impart toxic materials to the fluid stream. Filter manufacturers generally qualify filters through performance of standardized testing in compliance with compendial methods ***(16)***. These tests involve the introduction of filter extracts or actual filter samples into cell-culture systems. Reactivity of the cell culture to the test article is assessed visually, scored and compared against predefined acceptance criteria for toxicological safety. The results of these tests are commonly included as part of the filter validation guide.

Over the years there has been a shift away from animal-based testing, to better align with animal-welfare principles.

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### 4.4.2 Animal-Derived Materials

The filter manufacturer typically includes information on biocompatibility of animal-derived materials used in filter manufacture. The use of bovine tallow-derived stearates poses essentially no risk for transmission of transmissible spongiform encephalopathies (TSE) and other diseases. Typically, the manufacturing processes used to produce bovine tallow and tallow derivatives are very rigorous and meet or exceed safety requirements ***(17, 18)***.

## 4.5 Filter Cleanliness

Particulate contamination from the filter, filter hardware, filter installation and the process should be evaluated and considered since each source may contribute to particle burden in the product. Filter flush effluent samples are examined for particles greater than 10 microns and greater than 25 microns. Filter flush effluent is considered compliant upon meeting compendial guidelines for particulate matter in injections. Using these criteria, filters are also qualified as nonfiber-releasing ***(10-12)***.

In addition to particles, filters may be a source of other contaminants, such as endotoxin, organic carbon, or oxidizable substances. Potential sources may include surfactants, wetting agents, additives in the plastic component manufacture, manufacturing debris and materials of construction (MoC). Preflushing filters may reduce levels of particulates and contaminants and may be performed as part of the wetting process prior to integrity-testing. A filter can be preflushed based on the extractable studies, taking into account the sterilization method and process step.

**Note:** Filter flushing and preflushing are performed by the filter user.

## 4.6 Filter Extractables

Since sterilizing-filters are often used in the final manufacturing steps of a pharmaceutical production process, the effect of the filter on the final product should be evaluated. An extractable is any chemical component that is removed from a material by the application of an artificial or exaggerated condition (e.g., solvent, temperature, time). A leachable is a chemical component that migrates from a contact surface into a drug product or process fluid during storage or normal use conditions. These are typically a subset of extractables ***(19, 20)***.

Potential sources of extractables and leachables include, but are not limited to, membrane components (e.g., plasticizers, surfactants, antioxidants, residual solvents, device support layers), and device components (e.g., end caps, housings, cages, O-rings). Factors affecting leachables may include the chemical properties of the process stream, sterilization process, contact time, temperature, and volume-to-area ratios ***(21)***. Filtration of organic solutions may lead to higher levels of leachables than aqueous solutions.

Extractable data may be obtained from the filter manufacturer or may be generated by the filter user (see **Section 6.5** for filter user-generated data). Filter manufacturers can use model streams to determine the extractable profiles under worst case conditions ***(22)***. Although an important component of filter validation, the level of extractables and leachables from most sterilizing-grade filters that can potentially enter the process stream is relatively low. Flushing the filters prior to use can further reduce these levels as demonstrated by flush curves measured by TOC or other tests for specific leachables ***(21)***.

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Data should ideally include a comprehensive list of the compounds that could be leachables when the filter is used in the actual formulation under process conditions. The necessity to perform leachable study should be evaluated based on risk determined from extractable studies and risk to patient safety.

### 4.6.1 Chemical Compatibility

The chemical compatibility of the filter device must be evaluated to avoid potential filter damage or alteration and to avoid fluid contamination by either leachables or particulates. Filter manufacturers will have general chemical-compatibility guides based on the MoC and a range of standard chemicals. Since there may be numerous chemical interactions between filter device components and the process fluid or solvents, the typical chemical compatibility table provided by the filter manufacturer is commonly used as a starting point for further testing. More extensive testing should be performed by the end user to determine chemical compatibility of the filter device and process fluid under process conditions (see **Section 6.0** for more details).

Chemical-compatibility testing should encompass the entire device and depends on the fluid, filtration temperature, and contact time. Additional studies should be performed by the filter user. Additional product-specific studies may be conducted by the filter vendor on request from the filter user to ensure there is no impact to product CQAs. Common chemical-compatibility tests include integrity test, tensile strength, nonvolatile residue (NVR), extractables, particulates, flow rate, optical or scanning electron microscopy (SEM), burst pressure and membrane/O-ring thickness ***(23)***. Use of a combination of tests is recommended since a single test might not be able to detect subtle incompatibilities.

## 4.7 Operational Ranges

Manufacturers typically specify the maximum operational temperature, pressure, and sterilization temperature limits and provide water flow-rate data for their filters. Manufacturers provide maximum forward and reverse differential pressure limits at specific temperature conditions, incorporating adequate safety margins. This information may also reference limits at different temperatures, facilitating the selection of process flow rates, temperatures, and sterilization options compatible with the filter. **Section 8.0** addresses selection and validation of sterilization options by the filter user.

Filter-system flow rate at a given differential pressure is a function of membrane polymer and structure type, pore size, housing inlet and outlet diameter, effective filter area, fluid temperature and viscosity. During the design of a process filtration system, these variables should be evaluated for compatibility with the operating limits of the filters and the filter area selected. Filter suppliers also test pressure-pulsation cycles to determine the mechanical stability of the filter device. However, care should be taken to avoid pressures exceeding the manufacturer's stated specifications of maximum allowable differential pressure at specified temperature ranges.

**Note:** There may be a distinction between the operational ranges with which a filter is generally suitable, and the "validated filtration parameters" for a sterilizing filtration unit operation, as defined in process-specific bacterial retention study validation in order to cover scale-down worst-case process conditions (see **Section 6.0**).

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## 4.8 Shelf Life and Shipping Studies

Filter manufacturers will validate packaging and shipping of the filters according to industry standards to ensure that the products can withstand common transport hazards ***(24)***. In addition, filter manufacturers will establish shelf-life recommendations based on product stability under defined conditions, which will be valid if the filter is stored and handled properly. These recommendations can be used by the filter user to establish expiration dates or use-by dates.

Filters that are sterilized via gamma or x-ray irradiation will have expiration dates due to the sterility claim associated with the sterilization method.

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# 5.0 Filtration Process Design, Usage, and Handling Considerations

According to the FDA's *Guidance for Industry*: *Process Validation: General Principles and Practices (Revision 1)*, quality, efficacy, and safety must be built into the product and cannot be tested at the end of the manufacturing process. This concept is especially important for many sterilizing filtration operations, as the filter often represents the final microbial control barrier prior to patient dosing ***(25)***. This guidance document outlines three stages of process validation: Stage 1-Process Design, Stage 2-Process Qualification, and Stage 3-Continued Process Verification. **Section 5.0** focuses on the process design aspects, and **Section 6.0** focuses on the process-specific validation requirements for a sterilizing filtration process. Continued process verification is generally out of scope in this document and is covered in other guidance documents. The process design methodology discussed here is consistent with the principles of the International Council for Harmonization (ICH) *Quality Guideline (Q8): Pharmaceutical Development,* which is to first evaluate the design space, then develop a control strategy ***(8)***.

Membrane filters are used for multiple applications within the pharmaceutical industry and therefore, membrane filter types, designs and configurations are as varied as their applications. This chapter will focus on the process design and usage of sterilizing filters for the removal of microorganisms in pharmaceutical processing. Many of these design principles will also apply to the use of sterilizing-grade filters in locations other than the final sterilizing barrier for the control of bioburden, although those use cases have significantly reduced risk to product quality and may not require the same level of validation and controls to be applied ***(26)***.

## 5.1 Process Design: Design Space Evaluation

An initial evaluation of the design space for a sterilizing-filtration operation will generally encompass a survey of the product properties, compatibility and required operational ranges used in the process.

### 5.1.1 Product Properties and Filter Compatibility

A good first step in the design of a sterilizing-filtration process is to evaluate the solution that needs to be sterilized and how it will interact with the filtration system. A thorough evaluation of the properties of the filtrate will guide in the selection of a filtration membrane and housing system that can withstand chemical stress such as solvents, strongly basic or acidic media, and high-ionic strengths. After performing this type of analysis, it can allow the user to select filtration systems that are appropriate for the application.

For the second step of the compatibility analysis, an evaluation of the physical properties of the process fluid and intended processing parameters should be carried out. Characteristics such as the fluid temperature range, particulate load, and viscosity can also affect the filter membrane and filtration system selection. The

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system must be able to handle the mechanical stresses placed on it and remain integral for the duration of the filtration process. Since there may be numerous chemical interactions between filter device/SUS components and the process fluid or solvents, the typical chemical compatibility table provided by the filter manufacturer is commonly used as a starting point for filter selection.

Another dimension of filter compatibility to consider is adsorption of components from the process stream. Adsorption is a mechanism of product binding to the membrane or product contacting materials, which may affect the product composition and concentration. Potential adsorption of product components to the filter and other components of the filtration system (e.g., bags, connectors, tubing) must be considered independently. Considerations should include membrane type, MoC used throughout the filtration system, hardware, and support materials. Flow rate, product concentration, contact time, process fluid components, temperature and pH can also affect the level of adsorption.

In instances where elevated adsorption occurs in the beginning of the filtration process but levels off during filtration, the filtered volume could be pooled to achieve the specified product composition. During PD, adsorption tests are typically performed at small scale and confirmed at large scale. These tests can also be used to establish potential pretreatment options (e.g., buffer flush, soaking), necessary process parameter options or membrane polymer choices. Elevated levels of nonspecific product adsorption on the filter membrane can result in accelerated filter-fouling and performance degradation that would need to be studied as part of filter-sizing studies.

### 5.1.2 Defining the Design Space and Operational Ranges

Knowledge of the desired at-scale process prospectively allows setting ranges to be studied during the development of the filtration process. For example, a survey of the compounding process and the maximum and minimum volumes that will flow through the filter during the sterilizing filtration process can create an important set of operational ranges that the filter must be able to process. In addition, a subset of these parameters will define ranges that need to be validated later (see **Section 6.3**). While some parameters can be difficult to know during the initial PD exercise, it can be useful to evaluate the potential market demands for the product and maximum expected batch volume. This will potentially reduce the need for additional development and revalidation of the sterilizing-filtration process as the conditions change over the product lifecycle. **Table 5.1.2-1** shows a list of potential factors to consider during the process design phase that can help determine operational ranges to study.

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**Table 5.1.2-1 Factors to Consider When Developing Operational Ranges for Study During Sterilizing-Filtration Process Design and Development**

| Factor | Extent | Rationale |
|---|---|---|
| Temperature | Refrigerated to Warmed | Dependent on product quality effects and physical property optimization, can affect filter differential pressures and flow rate due to viscosity differences and also microbial growth rates |
| Total Product Volume | Minimum and Maximum | Sets limits for bacterial retention study validation and allows proper filter sizing |
| Flow Rate | Zero to Maximum Expected | Flow rate dependent on application, e.g., in line with sterile filling machine or bulk filtration |
| Differential Pressure | Minimum to Maximum | Requires appropriate development studies to determine pressure/flow rate relationship; reasonable maximum differential pressure should be defined based on expected conditions as well as a minimum expected to aid, for example, in detection of filter installation errors |
| Filtration Time | Maximum | Dependent on the processing time required, determines how long the process will take, including process setup and any necessary process stops |
| Product Contact Time | Maximum | Extended contact time may impact product compatibility, membrane fouling, or filter integrity |
| Driving Force | Pulsatile and/or Constant Pressure | Determines what type of pump or driving force will be used in the commercial setting; certain filters may be used in more than one manufacturing setup |
| Configuration | Single or Redundant Filter Setup or Multiple Filter Setup | Knowledge of physical size constraints for the filter may necessitate the use of multiple filters site, business or procedural factors may also require the use of redundant filtration; and knowing this up front will aid in the design of the filtration process |

## 5.2 Flow Characteristics

Flow-performance criteria during a filtration operation can include parameters such as flow rate, total throughput, thermal and mechanical resistance, and nonspecific adsorption. A filter's EFA is used in calculations for scaling filter performance, determining extractables levels, reporting flow rates (i.e., flux), and assuring of bacterial retention. Since applications vary, desired filter-performance characteristics also vary. For example, since buffers widely utilized in purification processes are nonfouling, the optimum filter may be one with the highest flow rate; for solutions that may contain foulants, such as media, the optimum filter may be one with the highest filtration capacity for that solution. Optimizing the filter to the process requires determination of critical and key operational parameters and performance requirements.

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Filter flow rate is affected by such parameters as the EFA, membrane porosity, pore size, thickness and area, differential pressure, flow path design, fluid viscosity, and temperature. **Table 5.2-1** shows the relationships between flow rate and a common list of process parameters.

**Table 5.2-1 Common Factors Affecting Flow Rate**

| Higher Flow Rate | Lower Flow Rate |
|---|---|
| High Porosity/Greater Void Volume | Low Porosity/Lesser Void Volume |
| Large Pore Size | Small Pore Size |
| Thin Membrane (less hydraulic resistance) | Thick Membrane (greater hydraulic resistance) |
| Large Effective Filtration Area (EFA) | Low Effective Filtration Area (EFA) |
| High Differential Pressure (fluid force) | Low Differential Pressure |
| Straight Flow Path | Torturous Flow Path |
| Low Viscosity | High Viscosity |
| High Temperature | Low Temperature |

Most of these parameters are governed by membrane and filter designs or particular fluid properties. The primary parameter is the differential pressure that is, the difference between inlet (feed side) and outlet (filtrate side) pressure. The differential pressure can be adjusted within the filter manufacturer supplied maximum temperature/pressure ranges to achieve the desired process flow rates. Studying the required pressure and flow rate ranges using scale-down membrane areas with the actual product stream is further elaborated in **Section 5.4.1.** It is important to understand this information thoroughly to determine the ranges that will be challenged during the validation activities (see **Section 6.8**).

### 5.2.1 Inlet and Differential Pressure

From the filtration curves it can be determined that increasing the pressure across the membrane will result in an increased flow rate under most processing conditions, which is an important variable to control during the sterilizing-filtration process. There are two important ways to specify the pressure: Differential pressure ($\Delta P$) is the difference in pressure between the upstream (inlet) and downstream (effluent) sides of the filter, while inlet pressure is the pressure measured directly on the inlet side of the filter. To determine the differential pressure, the inlet pressure must be measured using a pressure gauge or sensor. When appropriate, outlet pressure may be assumed to be atmospheric (e.g., a vented downstream vessel). If the outlet pressure is higher than the atmospheric pressure (e.g., when the fluid is transferred into a pressurized or unvented system or if the flow rate causes a significant pressure drop downstream of the filter) either the outlet pressure must be measured, or assume that the inlet pressure represents a worst-case upper limit on the true differential pressure.

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Depending on the flow regime and the safety factor to the validated maximum pressure it may be important to monitor the inlet or differential pressure of the filtration system (see **Section 6.8**). Increasing differential pressure at a constant flow rate is indicative of filter fouling, and detection may be important for certain types of process streams that are susceptible to fouling behavior.

Many modern filtration systems are driven by peristaltic pumps, especially in the case of SUS. As peristaltic pumps are a type of positive displacement pump, the flow rate can be easily controlled (correlated with pump revolutions per minute (RPM) and defined tubing diameter) without needing to worry about cleaning internal pump components or tubing systems; this adds a degree of safety and convenience to many filtration process designs. On the other hand, filtration can also be performed by pressurizing the system, managed by a pressure control valve, which avoids the need of a pump system. In most final sterilizing-filtration systems, the process stream is nonfouling and the filtration pressure remains relatively consistent over time. With proper system design and appropriate safety factors, constant flow rate or pressure filtration setups can operate in pressure regimes well below validated filter pressure limits. Some form of pressure control or monitoring may be an expectation of certain health authorities, even for pump-driven sterilization filtration processes that do not exhibit significant fouling.

### 5.2.2 Filtration Process Temperature

Process temperature has a variety of effects on the filter and fluid stream. Elevated temperatures reduce the maximum allowable differential pressure due to the strength of the mechanical structure of the filter. Pressure limits may vary between forward-and-reverse flow directions, based on filter structure (reverse flow often being considered worst-case). Pressure limits at specific process temperatures may be listed by the filter manufacturer in the qualification documentation of the particular filter type. **Figure 5.2.2-1** depicts an example of the relationship between process temperature and maximum differential pressure, that is, the higher the process temperature, the lower the pressure resistance of a filter. Under elevated temperature conditions, filter users should evaluate the risk factors, and if necessary, monitor process parameters carefully.

In addition to affecting differential-pressure capabilities, higher process temperatures could result in elevated leachables or chemical incompatibilities. Consideration should also be given to temperature, as some polymeric membranes are sensitive to hydrolytic and oxidative alterations by heat.

Higher temperatures reduce fluid viscosity and increase the flow rate through the filter for most Newtonian fluids. Higher viscosities could also result in the need for higher differential pressures to achieve a required flow rate. These elevated differential pressures correspondingly require increased monitoring of flow rate and/or differential pressure to avoid exceeding the rated pressure conditions.

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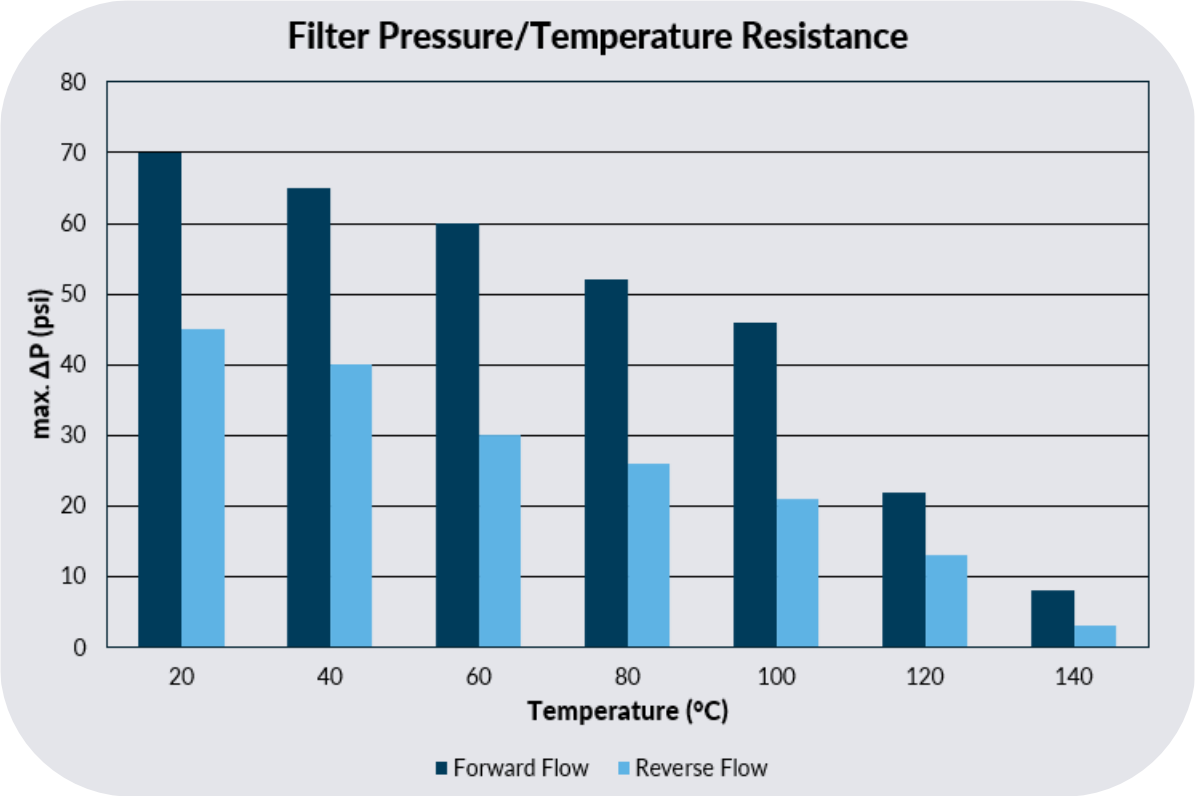


**Figure 5.2.2-1 Example of Filter Pressure/Temperature Resistance**

### 5.2.3 Filtration Time

Filtration time (duration) is the amount of time required to filter a single batch (or campaign) of production fluid. The process duration will differ depending on if the process is a single bulk-sterilizing filtration operation or an intermittent in-line-filtration operation used to maintain the level in a break tank connected to dosing pumps during a sterile filling process. Increasing filtration times may increase the potential for bacterial penetration; therefore, the filter retentive performance should be validated to the maximum product/filter contact time for the process (see **Section 6.8**).

As part of the development of the process an assessment should be conducted to determine the longest potential product-contact time with the sterilizing filter. The worst-case process time should include maximum filling-process duration, including manufacturing interruption scenarios prior to and during filtration.

## 5.3 Filter Throughput and Maximum Volume

Filter throughput, or capacity, is the measurement of a total volume that can be filtered through the EFA under defined process conditions. Filter capacity depends on membrane material, porosity, pore size, nonspecific adsorption, pore shape, filtration area, pressure differential, contaminant load, and the physical properties of the product type being filtered. **Table 5.3-1** lists potential factors affecting throughput.

Many process fluids that undergo sterilizing filtration are in the final formulation and do not result in significant fouling or filter-flux degradation over any practical volume. In many cases, there is no practical limit to the throughput, while in other cases, there can be particulate matter or subvisible particles present in

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the filtrate that do limit the filter throughput. In some cases, the prior processing history of the filtrate can affect the capacity (e.g., high-shear mixing of monoclonal antibodies can generate agglomerates) ***(27)***. Appropriate scale-down studies can be used to determine these behaviors as outlined in **Section 5.4**. Even in the case where no defined maximum filter capacity can be determined, the maximum volume that will be filtered should be specified so it can be covered during the filter-validation studies (see **Section 6.8**).

**Table 5.3-1 Factors Affecting Throughput**

| Higher Total Throughput | Lower Total Throughput |
|---|---|
| High Porosity/Higher Void Volume | Low Porosity/Lesser Void Volume |
| Large Pore Size | Small Pore Size |
| Low Nonspecific Adsorption | High Nonspecific Adsorption |
| Asymmetric Pore Shape | Isotropic Pore Shape |
| High Effective Filtration Area (EFA) | Low Effective Filtration Area (EFA) |
| Low Contaminant Load | High Contaminant Load |
| Nondeformable, Hard Contaminant | Deformable, Soft Contaminant |
| Low Shear-Stress History | High Shear-Stress History |

## 5.4 Scale-Down Systems and Testing

PD activities are generally done at the bench-scale using scale-down systems to minimize material quantities and allow for faster and more efficient experimentation. In order for the scale-down systems to be representative of the larger scales used in manufacturing the available systems and scaling considerations are discussed in this section.

### 5.4.1 Small Scale Device Testing

Filter-screening is generally performed using the process fluid in a constant flow or constant pressure experimental setup. Traditionally, 25mm or 47 mm discs or disc composites (**Figure 5.4.1-1**) installed in holders (**Figure 5.4.1-2**) were used for these screening studies. With modern filtration systems, however, most filter manufacturers have developed specialized encapsulated discs or small pleated capsules that better represent the flow dynamics of full-sized filters and allow closer estimates of scaling by EFA (see **Figure 4.2-2**). The screening tests can also be helpful to determine potential prefilter combinations to optimize the total throughput and reduce fouling, where required. For yield-sensitive applications, filtrate samples can be taken at frequent intervals to determine any nonspecific adsorption effects. Small-scale device tests may not always accurately predict the actual process in regard to flow-path dynamics and pleat-density parameters, however, and therefore may not scale up in a linear fashion, so care must be taken in the experimental design to ensure that any throughput or flux limitations are well understood.

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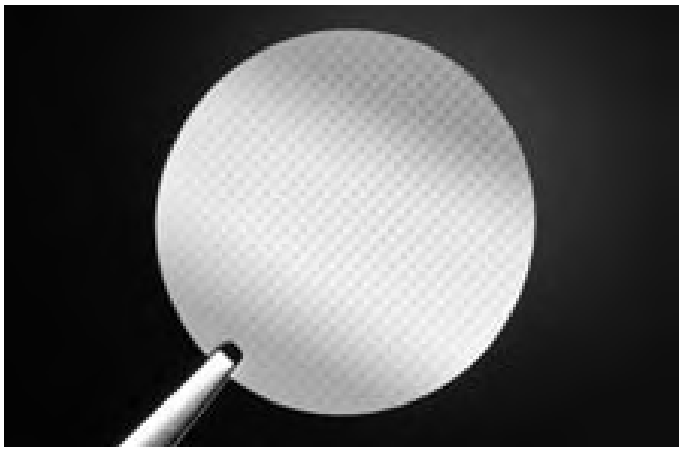

**Figure 5.4.1-1 Filter Disc**

**Figure 5.4.1-2 Filter Disc Holder**

Most filterability tests are performed by keeping pressure constant and measuring the decline in flow rate as a function of filtered volume. The pressure set-point should mimic production pressure conditions. Nevertheless, there are some applications that should be evaluated using constant flow rate, such as the filtration of media. For constant-flow testing, the flow rate is controlled while the pressure gradually increases until a maximum pressure is reached. This mode of testing avoids compression of contaminants that may be captured on the membrane surface, leading to membrane fouling. Either constant-flow or constant-pressure filterability tests can be used to predict manufacturing-scale performance. Ideally, small-scale tests should be performed using flux and pressure conditions representative of the manufacturing-scale process or a representative worst-case scenario.

The main output of these small-scale filtration studies is to develop and understand the flow curves and evaluate filter flux at different pressure conditions, thus allowing selection of the optimal filter size and configuration to use in the at-scale process. If properly designed, the fouling propensity of the filter and process fluid combinations can also determine a maximum filter-throughput value. With this information under representative temperature and process conditions, the process can be scaled up using the ratio of EFA.

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### 5.4.2 Filtration Process-Scaling Considerations

Scaling up filtration performance from a laboratory-scale filter unit to a production-scale system, which might contain multiple cartridges or units, poses several challenges to the filter user. Challenges include, but are not limited to:

- Product for filter trials might be in limited supply due to high production costs and limited production runs.
- Surrogate fluids might not be representative of the actual process fluid for certain behaviors like fouling or non-Newtonian flow dynamics.
- Small-scale filter studies may not represent the conditions under which filtration occurs in the manufacturing process.
- Laboratory-scale filter elements used for testing may not be directly predictive of large-scale filter capacity.

To overcome these challenges, filter users may elect to work with the filter manufacturer(s) to design small-scale studies that either closely approximate production systems or, alternatively, rely on appropriate safety factors to account for unknown variables and ensure successful filtration operations at scale.

Careful attention should be paid to ensuring that the material used for the scale-down studies is representative of the materials used for commercial production. In the production of proteins, there have been several documented examples where shear stresses imparted on the molecule upstream in the process result in the formation of subvisible particulates that can cause filter-fouling ***(27)***. In many cases the same solution that has not been subjected to the same levels of shear stress will show no signs of fouling behavior; this represents a scale-up risk that can be difficult to characterize that can manifest itself as filter-clogging during production. In this instance, considerable study may be required to create appropriate scale-down models of this shear behavior, as not all protein solutions and mixer types react in the same fashion. This is just one example of the importance of understanding the full-scale process and the stress history of the filtrate and how they affect filtration performance. There are also instances where freeze-thaw or shipping conditions of solutions, such as lipid nanoparticles (LNPs) or lentivirus drug products can affect the filtration performance (e.g., reduced volume throughput). It is critical that the impact of these factors is understood to ensure small-scale studies provide representative data.

## 5.5 Filtration System Design and Integration

A user requirement specification (URS) is a helpful tool in designing a SUS or stainless-steel setup. The filter user should have detailed knowledge of all process operational parameters to define the requirements of a particular assembly. The suitability of particular components depends on the requirements of the product and process. URSs typically include:

- Flow rate
- Pressure conditions
- Temperature conditions
- MoC
- Filter (pore size, area)
- Product volumes
- Instrumentation
- Sampling requirements
- System configuration
- Fluid properties
- Integrity-test methods and needs
- Sterilization method
- Documentation requirements

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In addition to the integrity-test considerations stated in **Section 7.0**, the following points should be considered for filtration systems:

- Pressurization of the upstream side of the filter with gas after wetting
- Need for a downstream vent filter to accommodate the gas flow
- Mechanism to aseptically remove the wetting fluid from the system, if necessary
- Use of a hydrophilic drain filter and a separate hydrophobic vent filter or other appropriate valves and drains for the system
- Use of a combination hydrophilic/hydrophobic vent filter, where applicable
- Appropriate placement of process controls, such as pressure sensors or flowmeters, and required sampling locations

**Note:** Use of separate filters enables both to be integrity-tested *in situ* with water, whereas a combination hydrophilic/hydrophobic filter can only be tested with alcohol, off-line.

The qualification tests listed in **Section 4.1** are performed within the manufacturer's defined conditions. Assemblies of components require validation within the filter/single-use user's process and environment. Validation should be performed to cover actual or worst-case process conditions, preferably with the product or a model (surrogate) solution. The single-use component or assembly should meet process and regulatory requirements. The validation effort should ensure that the components/assembly functions as intended and does not negatively influence product attributes, neither the product and process conditions, nor the single-use components or assembly.

Furthermore, functionality tests verify that desired process conditions, (e.g., filtration time) can be met. Documented evidence should be established that operational parameter specifications can be met repeatedly.

### 5.5.1 Multiple Filters in Series

Manufacturing processes are frequently designed with trains of multiple filters in a series (sequential flow through one filter after another), but the purpose of these extra filters varies from process to process, and the design and purpose of each filter must be clearly documented.

#### 5.5.1.1 Two-Filter Sterilization (Double Filtration)

In uncommon processes, where two filters together in series are **both** required to achieve sterility and the train has been validated to achieve the sterilization of a specific product only when passing through both filters, the filter train can be considered a single sterilizing unit, and both filters must successfully pass the integrity test.

#### 5.5.1.2 Pre-Filtration

One of the most common purposes for additional filters upstream of the final sterilizing filter is for prefiltration functions, as opposed to sterilization. An example is bioburden reduction that may be performed with a sterilizing-grade filter, but the filter is not performing a strict sterilizing function. Rather, it is assuring that a very low bioburden level is present before entering the final sterilizing filter.

Other prefilters may be performing functions unrelated to microbial control, such as aggregate or particle removal prior to the final sterilizing filtration, that may not use sterilizing-grade filters.

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### 5.5.1.3 Redundant Filtration

Processes where an additional (redundant) sterilizing filter is placed in the filter train to ensure against loss of product due to potential-integrity test failure of the primary sterilizing filter. The additional "secondary" filter is typically identical to the primary sterilizing filter, is fully validated to sterilize the fluid on its own and undergoes post-use integrity-testing as a contingency in the event the primary filter fails. Generally, the filter closest to filling (farthest downstream) should be the primary filter ***(10)***. The pressure and flow-rate controls are generally designed to apply across both sterilizing filters.

**Figure 5.5-1, Figure 5.5-2, Figure 5.5-3,** and **5.5-4** illustrate different filter trains that can be used for the filtration operation (see **Section 7.4.3** for additional information).

**Note:** DP/DS stands for drug product/drug substance.

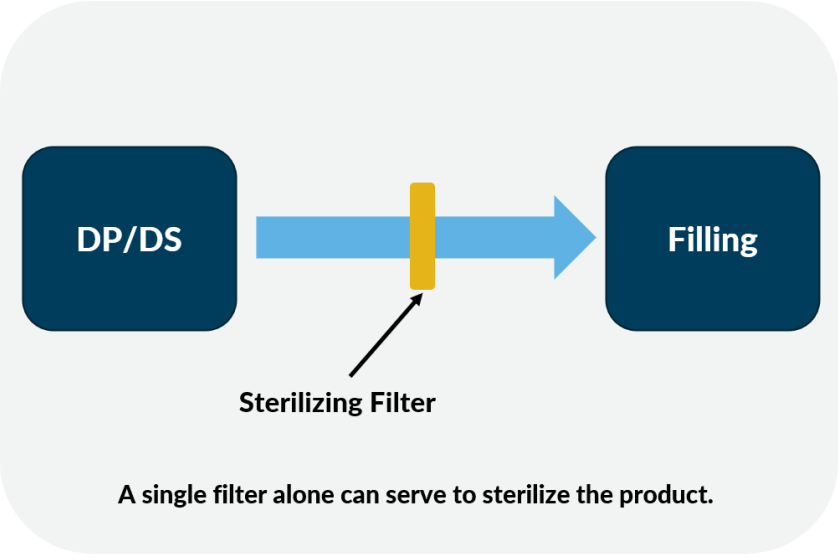


**Figure 5.5-1 Single Filter Train Configuration**

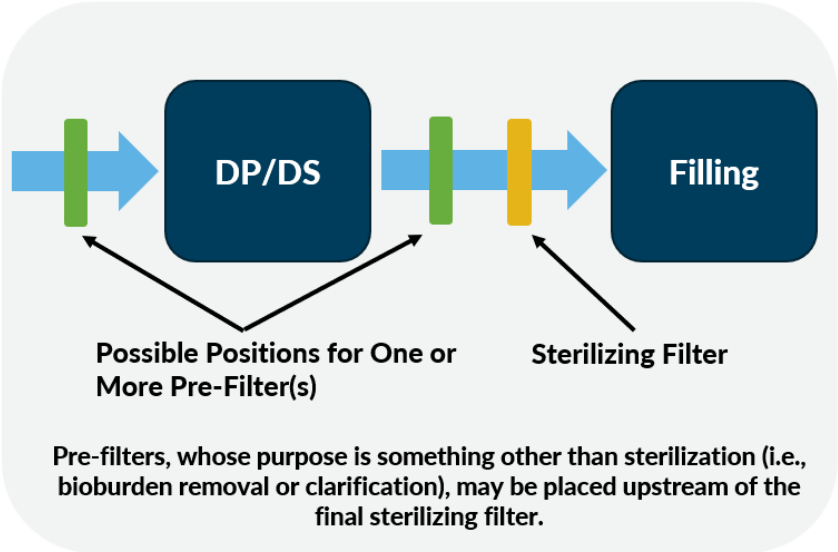


**Figure 5.5-2 Pre-Filter Train Configuration**

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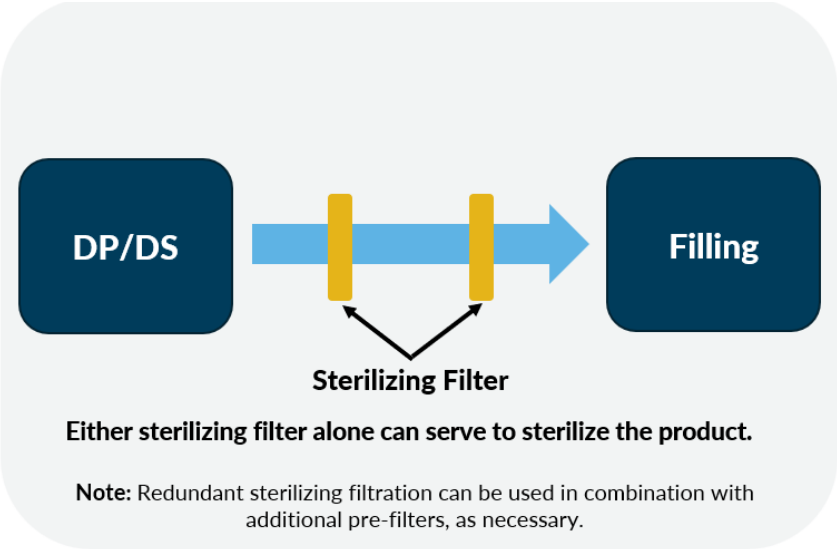


**Figure 5.5-3 Redundant Sterilizing Filter Train Configuration**

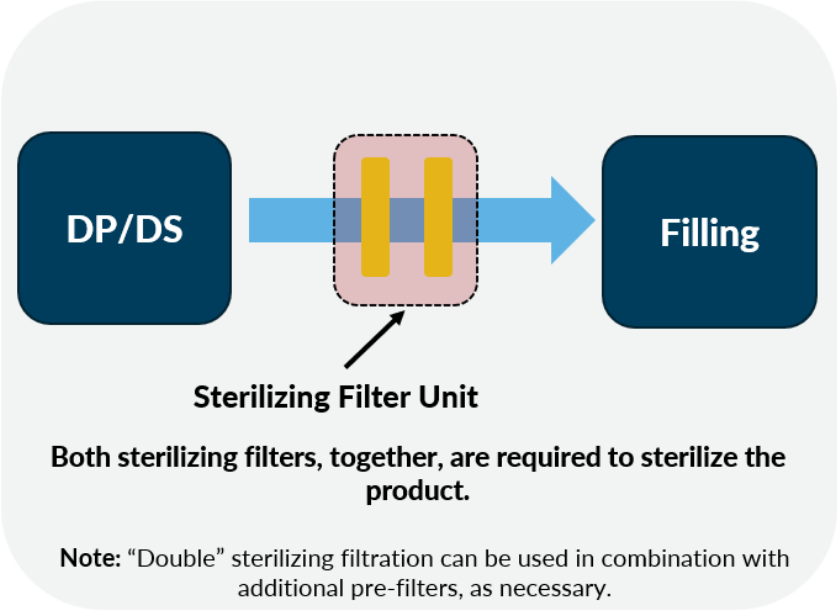


**Figure 5.5-4 "Double" Filtration Train Configuration (uncommon)**

## 5.6 Pre-Filtration Bioburden Monitoring

An important aspect of filtration process design is pre-filtration bioburden monitoring: a common regulatory expectation. The inlet to the final sterilizing filter should have well-controlled bioburden levels, despite the extreme bacterial removal efficiency typically validated in retention studies (see **Section 6.6**) ***(28)***. The difference between the low bioburden levels routinely entering the sterilizing filter, and the high levels validated to be removed by retention studies provides a large safety factor which is an important principle in sterilization by filtration. Factors to consider when designing a pre-filtration bioburden monitoring program include:

- Relevant regulatory expectations for allowable bioburden limit (i.e., the acceptance criteria of the test)

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- Representative sampling time:
  - If bioburden levels could rise over the course of the filtration operation, sampling at or near the end of the filtration could be warranted. For short filtration durations or bacteriostatic/bactericidal fluids, sampling at the beginning of the filtration may be appropriate.
- Location of the sample: the pre-filtration bioburden sample should be taken immediately upstream of the sterilizing filter (or, in cases of redundant filtration, upstream of the first sterilizing filter), and downstream of any bioburden reduction filters.
- Impact of preservatives on the bioburden assay.

## 5.7 Flushing Conditions/Filter Priming

Sterilizing-grade filters generally must be flushed and primed prior to use, either in the process of integrity-testing or as a separate step, and filter manufacturers typically have recommendations for minimum flush volumes. Pre-use flushing with water for injection (WFI), buffer, product, or sequential combinations as necessary may be used to reduce leachables and can be important to reduce the level of nonspecific adsorption and ensure that the filtrate is homogeneous during the process. Filter-priming is also necessary to remove air from the filters and ensure that the filter membrane is fully wet thus making use of all of the filter EFA during the process. For specific flushing considerations related to pre-use post-sterilization integrity test (PUPSIT) refer to **Section 7.0**.

Pre-use flush volumes are typically determined empirically at scale for each installation based on the needs of the process. Depending on the propensity of the filter to adsorb the product or excipients contained in the product stream, there may also be a specific flushing solution used. For example, polysorbate or poloxamer surfactants are common excipients in the final formulation of many protein-based pharmaceutical products. Depending on the filter chemistry and the formulation composition, the saturation of adsorption sites on the filter material with these excipients can occur considerably slower than the saturation of adsorption sites that chemically bind the protein active ingredient, leading to reduced excipient concentrations at the beginning of the filtration. In these instances, a surfactant containing flush buffer can considerably reduce the required product flush volume, significantly increasing the product yield. The flush steps can be estimated using a scale down system, although due to the flow dynamics and different surface area-to-volume ratios between small-and-large scales, they require confirmation at full scale. In these instances, a larger flush volume, calculated as a multiple of the EFA-ratio scale-down volume required, is used, the flush is done in a stepwise fashion, and the material is collected after the filter. This fractionated material can then be subjected to characterization testing, such as TOC, excipient concentration, or protein-specific assays that can be used to generate a filter-flush curve to determine when the leachables are removed, the filter reaches saturation equilibrium and the filtrate is completely homogeneous. Nonproduct flushes may need to be chased with a product flush to avoid diluting effects and return the filtrate to a homogeneous state.

## 5.8 Filter Handling and Installation

An important part of the development of a sterilizing filtration process involves the installation and handling of the sterilizing filters in the manufacturing environment. Filter cartridges must be installed into a filter housing. Filters encapsulated in a polymeric housing may be part of a pre-assembled SUS assembly. Most commonly, the filter cartridge will be installed into the filter housing and heat sterilized via in-line steam

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sterilization or autoclaving. Capsule filters are commonly supplied presterilized via irradiation, or they can be autoclaved by the end user.

### 5.8.1 Cartridge Handling and Installation Considerations

By following these guidelines for cartridge-handling during the development of the process, users can ensure the proper installation, maintenance, and performance of the filtration system.

1. **Identification and Receipt**
   - Cartridges should be identified by checking the part number, lot number, and description.
   - Enclosed documentation such as the quality certificate or certificate of conformance should be reviewed to verify the correct filter type and conformance with the specifications.
   - The outer and inner packaging material should be checked for damage with processes. put in place to segregate and evaluate any instances of damage.
2. **Installation**
   - Cartridges should be installed in the classified area as defined within the approved contamination control strategy (CCS) to mitigate contamination.
   - Cartridges should be handled with care and protected by the packaging during installation to avoid contamination or unnecessary operator contact.
   - Cartridges should be inserted into the filter-housing recess or adapter without twisting and ensuring a secure fit without forcing it in place. If the cartridge includes locking tabs, the cartridge should be rotated into position by holding it near the filter's base rather than at the closed end.
   - If needed, the manufacturer's recommendations for lubricating O-rings and using a product-compatible liquid should be followed.
   - Proper alignment and secure installation to prevent leaks or bypass.
   - Cartridges must be handled carefully to avoid damage to the filter media. Ensure proper procedures or instructions are in place to note any dropping or rough handling of the filtration system during installation.
3. **Wetting**
   - Some filter cartridges are required to be wet with water before steam sterilization or autoclave cycles. In that case care should be taken to achieve saturated steam conditions on the porous membrane structure during the sterilization phase and to ensure the differential pressure stays within the required conditions during the different sterilization phases.
   - Filter manufacturers typically have process recommendations on how to wet the filter to ensure the entire membrane matrix is properly wet.

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### 5.8.2 In-Line Steam Sterilization Considerations

In-line steam sterilization is one of the main sources of potential damage to a filter. Well-defined standard operating procedures (SOPs), a proper training program, and close qualification and monitoring of pressure conditions are essential to prevent damage during steam sterilization. In addition to the sterilization considerations outlined in **Section 8.0**, the following items should be considered during PD to ensure the filter is not subjected to undue stress during steam sterilization.

1. **Positioning**
   - During steam sterilization, the cartridge should be positioned to prevent the accumulation of condensate, allowing drainage from the housing.
2. **Temperature and Pressure Monitoring**
   - Pressure gauges should be installed as appropriate to monitor pressure conditions during the steam sterilization process or during qualification activities.
   - Excessive pressure conditions during steam sterilization can potentially damage the membrane of the filter cartridge.
3. **Potential Damage**
   - Reverse-steaming of the filter at elevated differential pressure conditions can cause damage to the filter, leading to issues like pleat-pack movement, cage imprinting, or cartridge deformation.
4. **Precautions**
   - Care should be taken to avoid condensation vacuum by pressurizing the system or venting through appropriate sterilizing-vent filters.

By following these guidelines for steam sterilization of filter cartridges, end users can ensure the integrity of the filtration system and ensure the effective sterilization of the cartridges for safe and reliable filtration processes.

### 5.8.3 Autoclaving Considerations

While the goals of steam sterilization using an autoclave are similar to those of in-line steaming, the process of placing a filtration device into an autoclave chamber is distinct in several aspects. Similar to sterilizing any item in an autoclave, the preparation of the filter must be considered, including appropriate wetting if necessary and how to wrap it to safeguard the sterilized equipment from contamination when removed after the sterilization cycle. The wrapping material must strike a balance; it should be porous enough to allow the penetration of steam and air while effectively blocking any bacterial ingress. Autoclave cycles typically run a pre- and post-vacuum cycle to evacuate any residual air and to dry the load. These vacuum cycles can cause high differential pressure when the wrapping materials are affixed too tight on the inlet, outlet and vent connections. Sterilizing-grade filters are vulnerable to damage from excessive differential pressure across the membrane at elevated temperatures; therefore the preparation of the filter requires attention during the development phase.

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In addition, the placement of the filter within the autoclave is of importance as it should not have any contact with other parts within the autoclave especially the autoclave walls as this could result in temperature gradients which can result in additional stress on the filter.

### 5.8.4 Filter-Capsule Handling

The filter-capsule handling can be compared with the filter-cartridge activities, except the filter capsule is a filter unit that does not need to be installed into a filter-housing; it is already encapsulated and just needs to be connected into the process stream. More detailed information on the handling of single-use capsule filters can be found in PDA TR 66.

Most filter capsules cannot be in-line steam-sterilized and either require to be autoclaved or are supplied presterilized using gamma or x-ray irradiation. The autoclaving cycle for filter capsules requires the same attention as the filter cartridge to prevent damage to the filtration system

## 5.9 Impact of Filter Sterilization

As shown in **Table 4.1-1**, the effect of the sterilization method and conditions on filter bacterial retention for both the membrane and the assembled filter unit should be a part of the filter manufacturer's qualification package. These studies are generally performed to demonstrate that stresses that occur during the sterilization process do not affect the ability of the filtration system to retain microbes. The work that the filter manufacturers perform demonstrates, that after sterilization stress, the filter performs as designed as long as the integrity test passes. Within the sterilization stress ranges studied, this allows nondestructive integrity-testing of the filters by the end user to demonstrate that sterilizing filters are functioning as designed. It should be noted that these sterilization-method qualification studies are different than those performed for product-specific bacterial retention validation studies, as outlined in **Section 6.0**. In the absence of any filtration surface modifications caused by the sterilization method, no additional product specific studies related to the filter sterilization conditions are generally required.

As long as the sterilization processes performed by the end user are within the ranges (e.g., temperature, time, ionizing radiation exposure) qualified by the filter manufacturer, performing an integrity test is an appropriate control strategy that connects the filter performance back to the microbial-retention testing performed during the filter qualifications.

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# 6.0 Product and Process-Specific Validation Testing for Sterilizing Filtration

Process-specific filter validation studies, to provide assurance that the filter will perform as expected under worst-case processing conditions, are required prior to submission for marketing authorization. However, filter validation studies may be conducted earlier based on risk assessment. A risk assessment can be based on prior knowledge from how the product, process and filter interact, leveraging successful filter validations of similar molecules. This risk assessment can be performed to evaluate whether a product operates within the design space. Bacterial retention studies, with the final sterilizing-grade filter, are required when a sterile claim is made and the product is not terminally sterilized ***(29)***. This section describes factors to consider when performing process-specific filter validation studies.

## 6.1 Performance of Process-Specific Filter Validation

The rationale for selection of a suitable filter for final sterilizing filtration should be documented in an end user's CCS. As described in **Section 5.0**, during PD, end users will review generic information available from their supplier (i.e., qualification data) to assess suitability of components for the intended process and identify any potential risks associated with compatibility, adsorption, and extractables as part of their initial risk assessment. Consideration should also be given if the final sterilizing filter will be part of a SUS, which could include tubing and bags, as this may impact the maximum processing parameters and potentially the design of the final process-specific filter validation studies. While it is recognized that process-specific filter validation studies must be performed prior to commercial manufacture or release of product, there is no specific guidance that states precisely when such studies should be performed in the development/validation lifecycle. Consequently, the decision when to perform filter validation is determined largely by the end user. This usually coincides when the drug product is in late phase development, as it is important that the proposed critical process parameters (CPPs) and full-scale manufacturing conditions are well understood to ensure the testing performed validates the intended filtration design space. If a specific risk is identified, it may be appropriate to conduct process-specific testing earlier during PD (e.g., if the drug is subject to expedited clinical development). Generation of early process-specific data can also be considered for the purposes of risk assessment and not the final filter validation.

## 6.2 Purpose of Process-Specific Filter Validation Testing

The purpose of performing process-specific validation testing for final sterilizing filtration is twofold:

1. To assess any impact the filter may have on the quality of the final drug product.
2. To assess any impact the drug product may have on the performance of the filter.

**Table 6.2-1** provides a summary of typical testing that is in scope for process-specific filter validation.

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**Table 6.2-1 Test Types for Process-Specific Filter Validation**

| Objective | Test Type | Purpose |
|---|---|---|
| **Determine Any Impact of Filter on Product*** | Adsorption | Determine any loss of product components to the filter membrane and components. |
| | Extractables | Assess potential compounds released from the filter into the final product under exaggerated conditions ***(30)***. If documentation is adequate these tests do not need to be repeated (see **Section 4.6**). |
| | Leachables | Assess potential compounds released from the filter (and) into the final product under normal operating conditions. |
| **Determine Any Impact of Product on Filter Performance (e.g., bacterial retention)** | Viability, Inhibition and Bacterial Retention Study | Determine method for performing bacterial retention study based on properties of process fluid.<br>Demonstrate bacterial retention of *B. diminuta* (or other appropriate process isolate) at a minimum challenge level of ≥ 1 x $10^7$ CFU/$cm^2$ under simulated worst-case processing parameters. |
| | Chemical Compatibility Filter | Confirm filter is compatible with process fluid / operating parameters and will perform as expected. |
| | Product Wet Integrity Test Value Determination (optional – not a regulatory requirement) | Define integrity test limits for a sterilizing-grade filter using the process fluid (or alternative fluid) that are correlated to bacterial retention. |
| *Any impact of the filter on the product CQAs should be conducted by the end-user during PD. Refer to **Table 4.1.1.** | | |

## 6.3 Selection of Worst-Case Test Parameters Following Quality-by-Design Principles

The principles of quality by design (QbD) should be applied when considering the worst-case parameters and conditions in which the sterilizing-grade filter should be validated. This includes a rationale for selection of worst-case parameters that must be based on a thorough understanding of the intended design space, including the normal operating range (NOR) and proven acceptable range (PAR) ***(8, 31)***. While it is expected that an end user will apply a safety factor to validate the use of the final sterilizing-grade filter in the given application, this should represent reasonable control of a process (based on manufacturing control

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limits) and defines the PAR for use of the sterilizing-grade filter in the application as outlined in **Section 5.0**.

A summary of worst-case parameters is included under the description of factors to consider for each filter validation test type.

## 6.4 Adsorption

While adsorption studies are typically performed during PD, as described in **Section 5.0**, some end users elect to have this performed as part of the filter validation package. Considerations for worst-case test conditions to assess potential adsorption are shown in **Table 6.4-1**, but the parameters selected for testing should be based on risk assessment.

**Table 6.4-1 Considerations for Worst-Case Test Parameters to Assess Potential Adsorption**

| Description | Worst Case | Rationale |
|---|---|---|
| Flow Rate | Case Dependent | Adsorption may occur more rapidly at lower flow rates but should be based on risk assessment |
| Contact Time | Shortest | Shortest contact time may not allow filter membrane to become saturated |
| | Longest | Extended period of static soak prior to filtration may not allow filter membrane to become saturated |
| Temperature | Dependent on Product Composition | Most adsorption processes are inversely related to temperature ***(32)*** |
| Product Component Concentration | Consider Preservatives, Surfactants, Active Pharmaceutical Ingredient (API) | Increased potential for adsorption which may impact product quality control specifications |
| Component Concentration | Lower | Increased potential for adsorption due to ratio of product concentration: membrane/surface area |
| pH | Test Extremes | May impact surface charge of membrane |

## 6.5 Extractables and Leachables

During filter selection, extractables (and potentially leachables, based on a risk assessment of the extractables data) should be considered during early-PD. This is typically assessed based on the availability of supplier data using standardized conditions (e.g., model solvents under specified test conditions) ***(22, 33, 34)***. As defined in **Section 4.6,** an extractable is any chemical component that is removed from a material by the application of an artificial or exaggerated condition (e.g., solvent, temperature, time). A leachable is typically a chemical component that migrates from a contact surface into a drug product or process fluid

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during storage or normal-use conditions. However, on occasion, interaction with product components may produce leachables not seen as extractables ***(20)***.

Extractables testing may be repeated as part of the final filter validation. After selection of a suitable extraction solution(s) (model solvent, surrogate or combination of solutions), extraction studies should be designed to encompass the worst-case process conditions with regard to key variables, such as temperature, time, pH, and pretreatment steps (e.g., flushing, sterilization) (see **Table 6.5-1**). Extraction studies are performed with the filter device using worst-case conditions of contact time and temperature with filters that have been sterilized by autoclaving or irradiation. Extraction studies may be conducted using methods such as static soak or recirculation/reciprocation/agitation. For static soak, the filter is submerged in the extraction solution at a given temperature for a defined amount of time. Extractables may also be generated by recirculating or reciprocating the extraction solution through the filter for a given amount of time under a defined temperature. The extract is collected and tested for the presence of filter extractables.

**Table 6.5-1 Worst-Case Parameters for Extractables/Leachables Testing Selection Factors**

| Description | Parameters | Worst-Case | Rationale |
|---|---|---|---|
| **Process Fluid** | Product Type | Solvents | Increased potential for extractables |
| | Product Type | Process Fluid | Evaluate leachables under normal processing conditions (if required) |
| | pH | pH Extremes | Increased extraction potential at extreme pH values |
| **Process Conditions** | Temperature | Higher | Higher temperatures may generate more extractables |
| | Duration | Longer | Longer durations may increase potential extractables |
| | Filter Sterilization | Higher | Higher autoclave temperature and/or cycles increases potential for extractables<br>Higher irradiation dose may increase potential for extractables |

Once the filter extracts are obtained, a quantitative and qualitative assessment of the filter-derived materials are determined through selection of appropriate analytical methods. Additional information regarding these methods can be found in BioPhorum's *Disposables: Extractables Testing of Polymeric Single-Use Components Used in Biopharmacetuical Manufacturing* ***(30)***. In addition to determining identities and quantities of filter extractables, patient safety is evaluated through a toxicological assessment.

Based on a qualitative (e.g., extractable chemical name with CAS number) and quantitative assessment of extractables to evaluate the patient exposure to potential leachables, the end user may also elect to perform a

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leachables study. This involves using the actual process fluid with the same filter type that will be used in production. Due to the low levels of potential leachables, extractables testing using a model solvent approach is usually a prerequisite to having a leachables study performed. Circumstances, such as drug product interference with analytical methods or the prohibitive cost of the drug product, may necessitate the use of a surrogate solution. Surrogates should closely resemble the product to be filtered. Another option is to use several solutions to bracket the properties of pH, ionic strength, or level of organic components of the actual fluid. When a surrogate or bracketing approach is used, a clear rationale for the selection of solutions and extraction conditions should be documented.

## 6.6 Bacterial Retention Study

Bacterial retention studies are used to validate a filter to perform the sterilizing function (i.e., render the filtrate sterile) by showing complete microbial removal from the challenge fluid, product, or product family, using a representative challenge microorganism ***(1, 35, 36)***. Two primary elements are taken into consideration for bacterial retention study validation:

1. Classification of the filter membranes as sterilizing-grade by means of bacterial retention studies , using standardized test methods ***(6)*** . This is the responsibility of the filter supplier as part of the filter qualification to demonstrate that the filter retains bacteria using standardized methodology (i.e., proof that the filter retains *B. diminuta* when challenged at a level of $\geq 1 \times 10^7$ CFU/cm$^2$ and can be classified as a sterilizing-grade filter in accordance with the principles of ASTM F838).
2. Demonstration by the filter user or designated test facility (e.g., filter manufacturer or contract laboratory) of complete retention of a microbial challenge from each product or product family in the actual process conditions. Scientific rationale for each product group should be developed and may be reviewed with the appropriate regulatory agency before validation testing. It is the responsibility of the end user to confirm that the selected sterilizing-grade filter performs as expected for process-specific applications (i.e., under simulated worst-case processing conditions with the intended drug product).

These two filter-testing concepts are not interchangeable and should be validated independently.

### 6.6.1 Factors Influencing Bacterial Retention

Factors potentially affecting bacterial retention include, but are not limited to ***(2, 3)***:

- Filter type (e.g., asymmetric, isometric)
  - Base polymer, surface modification chemistry, pore size distribution, and thickness
- Fluid components
  - Formulation, surfactants, and additives
- Fluid properties
  - pH, viscosity, osmolarity, ionic strength, and surface tension
- Process conditions
  - Filter reuse, temperature, pressure differential, flow rate, filtration time, total filter/product contact time, and volume throughput.* The potential impact of intermittency should be considered.

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- Potential bioburden in the product (quantitative and qualitative)

***Note:** As sterilization of the filter is part of the supplier qualification performed during development, exact simulation of the end user's sterilization method (e.g., irradiation, autoclave, SIP temperature, number of cycles) is not generally required for process-specific bacterial retention study.

In addition to these factors, process bioburden should be evaluated in order to establish *B. diminuta* as a relevant organism. Evaluation should be based on bioburden identification and a risk assessment to ensure the absence of potentially more penetrative organisms in the prefiltration bioburden (see **Section 6.6.2**). Consideration should be given to potential product formulations or process conditions affecting cell size or other physiological or morphological microorganism attributes that might allow organism passage.

### 6.6.2 Challenge Organism Selection Criteria

Historically, *Pseudomonas diminuta*, reclassified as *B. diminuta ,* has been the microorganism of choice for bacterial retention studies. Although a product may be bactericidal to *B. diminuta* under normal processing conditions, other microorganisms may survive under the same conditions. Another bacterial retention study method for bactericidal products is the use of "indigenous bioburden." Indigenous bioburden consists of bacterial isolates from the manufacturing environment or the product formulation that have demonstrated the ability to survive within the product formulation under actual production processing conditions.

In process bacteria should be propagated or equilibrated in the product in order to ensure that their morphological and physiological characteristics are representative of process isolates. The smallest, or worst -case, indigenous organism should be used based on a risk assessment against use of *B. diminuta* as the challenge organism. For additional discussion of preparation and use of challenge organisms, see **Section 6.6.3.**

If diminutive microorganisms are known to exist in the product, they or an appropriate model organism, may be used to challenge test filters. Typically, these challenges are conducted in addition to *B. diminuta* testing, and it is strongly recommended that appropriate measures be taken to remove these organisms as a potential contaminant.

If other bacteria are used, they should be small enough to challenge the retentivity of the sterilizing-grade filter and should simulate the smallest microorganism found in production ***(4)***. A risk assessment should determine the likelihood of any prefiltration bacteria potentially being able to penetrate a 0.2 µm sterilizing-grade filter ***(37)***.

The size of the challenge microorganism should be confirmed by demonstrating passage through a 0.45 µm-rated membrane as a positive control for each challenge performed. This confirms the challenge organism is appropriately sized and monodispersed. *B. diminuta* grown under standard culture conditions penetrates 0.45 µm-rated membranes in small numbers at high challenge levels, typically $\geq 10^7$ CFU/cm$^2$ (see **Section 6.6.3**). In some cases, *B. diminuta* may not be a representative worst-case model. Documented rationale should be provided if a different challenge organism is chosen.

### 6.6.3 Culture Maintenance and Challenge Preparation

*B. diminuta ATCC® 19146* ™ can be obtained in lyophilized form from bacterial culture collections such as the American Type Culture Collection (ATCC). After reconstituting the microorganism according to the manufacturer's instructions, stocks can be maintained (either refrigerated or frozen) on appropriate media

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per standard microbiological practice. Maintenance conditions for process isolates to be used for challenge studies should be established.

Two standard techniques have been recognized as suitable for preparation and maintenance of *B. diminuta* for bacterial retention studies. These are the SLB and the frozen cell paste (FCP) methods. Both methods have been found to be effective in producing suitable suspensions of *B. diminuta* of approximately 0.3 – 0.4 µm in diameter by 0.6 – 1.0 µm in length.

Alternate media and culture methods may be equally valid to prepare *B. diminuta* provided they produce single, monodispersed cells suitably sized to penetrate 0.45 µm-rated membrane filters. Alternate culture methods should be validated. Bacterial retention study stock cultures can be screened for aggregation by optical microscopy. In cases where aggregation is observed, immersion of the stock culture in an ultrasonic cleaning bath filled with cold water for a qualified duration is one means of dispersing the aggregate. The cavitation action of the bath is effective in disaggregating bacterial cells without the loss of viability. This effect should be confirmed by optical microscopy, viable count, and recovery downstream of the 0.45 µm-rated size control filter ***(38)***.

The bacterial challenge concentration should provide a uniform challenge over the duration in which viability of *B. diminuta* is assured to yield a final challenge level of at least $10^7$ CFU/cm$^2$ of filter surface area. Operational parameters (e.g., flow rate, time, and pressure) should be taken into consideration when calculating the bacterial retention concentration.

A bacterial challenge level of ≥ 1 x $10^7$ CFU/cm$^2$ is, therefore, the requirement for a sterilizing-grade filter (historically a filter rated at 0.2 µm). This level was derived from Bowman's observations that *B. diminuta* could penetrate a 0.45 µm-rated membrane at > $10^4$–$10^6$ CFU/cm² bacterial challenge level, and her suggestion for qualifying 0.2 µm-rated membranes as "sterilizing-grade" with *B. diminuta* at $10^7$ CFU/cm² to assure minimally sufficient sensitivity to detect oversized pores ***(39)***. Bacterial challenge concentration (CFU/mL) should not be confused with bacterial challenge level (CFU/cm$^2$).

The viability and titer of the *B. diminuta* suspension should be confirmed using a suitable recovery medium such as soybean-casein digest agar (also known as tryptic soy agar), Mueller Hinton Agar, or other qualified growth medium. When performing filter challenges, the viable titer of the bacterial retention suspension should be determined immediately prior to and after the challenge; the bacterial titer should be determined using an accepted microbiological testing method. For situations when this is not possible (e.g., extremely short-duration bacterial retention studies), a rationale should be provided to justify the use of either a pre- or post-input viability sample. The same culture medium should also be used to determine any recovery of *B. diminuta* downstream of the test and control filters.

### 6.6.4 Viability/Filter Flush Qualification Testing Procedure and Protocol Development

Prior to completing the bacterial retention study, the test organism's viability should be verified by direct inoculation into the carrier fluid (ideally, product or surrogate fluid, if justified) in order to determine the appropriate challenge test methodology (see **Figure 6.9-1**). Scientific justification should be provided if viability testing is not performed before finalizing the bacterial retention study design (e.g., nonaqueous process fluids). The microorganism should be grown in the same manner as that used for challenge-testing to preserve its morphological and physiological characteristics. The test exposure time chosen for the viability studies should equal or exceed the maximum filter/product contact time, unless viability of the test organism is demonstrated to decline by ≥ 1 log before the actual process filtration time is reached.

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**Note:** It is not necessary to determine the exact point in time at which viability of the test organism declines by ≥1 log; however, the viability test time should be sufficient to allow meaningful measurements to be taken during the bacterial retention study.

The organism should be added to the process fluid (or surrogate solution) such that the inoculation volume does not cause significant dilution of the drug product and viability should be monitored by taking appropriate samples over the intended process duration to determine cell concentration in accordance with standard methods. A control (e.g., water, buffer) should be spiked at the same ratio and viability of the organism monitored over the same duration. If the temperature may be inhibitory to the viability of the challenge organism (typically, over 40ºC for *B. diminuta*), consideration should be given to assessing viability at ambient temperature so the bacterial retention study could be conducted by inoculating the organism directly into the process fluid ***(40)***. In addition, for broad processing temperature ranges, consideration may be given to the impact of the extremes of the temperature range on viability of the challenge organism. If there is less than an approximately 1 log difference between the test and control over at least 30 minutes, direct inoculation of the organism into the process fluid may be an option for the bacterial retention study. Additional consideration should be given for situations where the viability of the challenge organism in the process fluid is reduced by close to 1 log at the desired time point. For such cases, it may be necessary to reduce the total exposure time of the challenge organism in the process fluid to reduce the likelihood of die-off, which could prevent the minimum challenge level being met during the bacterial retention study. A decision should be made between ensuring the minimum challenge level of $\geq 1 \times 10^7$ CFU/cm$^2$ is met and ensuring the bacterial retention study duration is sufficiently long to allow adequate measurements of differential pressure, flow rate, volume throughput and temperature.

Filter flush schemes should also be considered prior to performing a process-specific bacterial retention study. If the drug product is bactericidal to the challenge organism, then adequate removal/neutralization of bactericidal effects resulting from residual process fluid should be ensured prior to introduction of the bacteria to the test system. If the process fluid is nonbactericidal and used as the challenge vehicle, the lack of potential inhibitory effects to the challenge organism should be ensured during the incubation period.

After the test organism's viability within the product has been established, the challenge methodology and protocol should be developed.

### 6.6.5 Bacterial Retention Validation Studies

The goal of bacterial retention validation studies is to have documented evidence demonstrating that the filtration process will consistently remove a high level of a standard bacterium, or relevant bioburden isolate, suspended within the product or surrogate fluid under simulated process conditions.

The decision to test membrane discs or a full-size process filter depends upon the study goal. Ideally, the test filter should mimic the process filter unit as closely as possible, including the membrane type and pore size. If the study is to validate the bacterial retention efficiency of a particular membrane material, the use of a small test membrane disc is generally regarded as sufficient, but the filter selected should be justified. The test methodology used for determining physical integrity of process filters should yield results that are meaningful in terms of bacterial retention studies, and correlation data should be available from the filter supplier. If different test methods are used, a relationship between the two should be demonstrated.

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The following should be considered when conducting process-specific bacterial retention validation studies on membrane filters:

- A thorough assessment of the filtration process should be conducted, including the nature of the process fluid (e.g., aqueous, acid, base, organic), active filtration volume, total product/filter contact time, process pressure, process flow rate, process temperature, number and/or frequency of intermittent cycles ("on/off"), and process-filter design specifications.
- Multiple filter membrane lots (typically three) should be included in process-specific bacterial retention validation studies. In cases where the product may be considered aggressive to the membrane, the exact number of filters and the test design will depend on the process.
- At least one of the three filter membrane lots used for the process-specific bacterial retention validation study should have a pre-study or pre-use physical integrity-test value as close as possible to the filter manufacturer's test specification. The filter manufacturer is responsible for defining the criteria for the minimum specification filter. Due to improved process control during membrane manufacturing, it may not be possible to obtain membranes at or near the filter manufacturer's production limit. In this case, test membranes should be selected from a manufactured lot whose pre-use integrity test values are as low as practical but representative of the standard filter-manufacturing process.
- Membrane physical-integrity test values from bacterial retention validation studies should be included in test reports. The physical integrity should be determined prior to and after challenge-testing using water, product, or other wetting fluid for which specifications exist. Use of a low-surface-tension wetting fluid for post-use integrity-testing may be required to remove residual product.
- An investigation should be conducted if the test organism is recovered downstream of any filter after the product bacterial retention study. If the investigation confirms penetration of the filter by the test organism and the filter meets its integrity-test specification, then the applicability of this filter under these process conditions should be reconsidered.
- Families of products with the same ingredients, varying only in concentration, may be validated by challenging the concentration extremes and accepting the intermediate concentrations by bracketing. If a single product is determined to be a worst-case representative, then rationale and data should accompany the model. A placebo (without the API but belonging to the same product family) may be considered for product bracketing with appropriate justification.
- During bacterial retention studies, incorporation of pressure increases to simulate integrity-testing (e.g., PUPSIT) should be based on risk assessment performed by the end-user (e.g., if process fluid is used for PUPSIT that could potentially pose a risk to bacterial penetration). Examples of factors to consider for risk assessment include the potential bioburden in the wetting fluid (based on a qualitative and quantitative assessment), type of integrity-test performed, and maximum number of integrity-tests that the filter could be exposed to. Filter suppliers qualify the filter to be able to withstand the elevated pressures resulting from integrity-testing. Therefore, the risk assessment should be focused on the potential for any bioburden in the wetting fluid to penetrate the filter as a result of PUPSIT.
- Under typical manufacturing conditions with controlled conditions, the risk of bacterial penetration due to PUPSIT is considered low. If PUPSIT simulation is deemed necessary, the test design should be accompanied by a technical rationale.
- Filter reuse is typically not practical or recommended for pharmaceutical purposes. However, if a sterilizing-grade filter is reused, justification should be provided, and reuse parameters should be validated. Scale-down laboratory studies are unlikely to be able to simulate the conditions of filter reuse by an end user (e.g., multiple sterilization cycles, clean in place, excessive handling). Therefore, consideration should be given to testing process filters that have been exposed to the entire cycle of reuse by the end-user to supplement the small-scale study.

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### 6.6.6 Bacterial Retention Validation Studies: Considerations for Selection of Worst-Case Test Conditions

There are different levels of risk associated with process-filtration parameters. Some are associated with microbial proliferation in the product prior to filtration, while others are associated with a higher risk of passage of bacteria through the filter. While some generalizations can be made with respect to selection of worst-case test parameters, it is important that worst-case parameters are determined on a case-by-case basis. Further, the worst-case parameters should be selected such that the intended manufacturing conditions are covered by the process-specific bacterial retention studies but should be reflective of the processing parameters established during PD (with an appropriate and justified safety factor). Bacterial challenge over the full process time assesses the potential for time-dependent factors, such as filter compatibility, maintenance of integrity and occurrence of time-dependent penetration.

The pressure differential across the test filters demonstrated during bacterial retention studies should meet or exceed the maximum pressure differential permitted during processing, within the filter manufacturer's design specifications. The intended process flow rates should be incorporated when designing the challenge conditions. It may not be possible to maintain both the target pressure and flux for the full duration of a retention study, while also achieving the desired total filter wet time and (scaled) batch size. In such cases, static wet hold times, recirculation, or intermittent filtration, can be incorporated into the study design when justified.

Bacterial retention study conditions should simulate the production process as closely as possible, but variations may be acceptable when they are well-justified and unlikely to affect the risk of bacterial penetration. Since bacterial retention studies are generally performed in a laboratory, the methodology should be scaled accordingly. The flux should be scaled to an equivalent flow rate per unit area, expressed for example in (ml/min)/$cm^2$. If filtration is regulated by pressure, the challenge test pressure should represent the NOR with consideration to the maximum filtration pressure allowed within the process and the likelihood of operating the filtration process at the PAR. Product recovery by blowdown may be performed at a higher pressure than maximum pressure during normal filtration and should be simulated at the end of the bacterial retention study.

**Table 6.6.6-1** lists examples of factors to consider when selecting worst-case parameters for process-specific bacterial retention studies.

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**Table 6.6.6-1 Considerations for Selection of Bacterial Retention Study Parameters**

| Description | Parameters | Worst-Case | Rationale |
|---|---|---|---|
| **Drug Product Properties** | Viscosity | Higher | Potential to cause higher pressure to the filter |
| | Surface Tension | Lower | Potential to contribute to bacterial penetration (based on higher-risk fluids, such as liposomes) |
| | Osmolarity | Higher | Potential for reduction in cell size over prolonged exposure |
| | Components | Liposomes, Emulsions with Lipids | Potential to contribute to bacterial penetration |
| | Bioburden | Diminutive Organisms | Potential to contribute to bacterial penetration |
| | pH | Neutral | Neutral pH more likely to support bioburden |
| | Ionic Strength | Consider Extremes | Potential impact to adsorptive retention |
| **Process Conditions** | Temperature | Case-specific | Lower temperatures may increase product viscosity leading to increased pressure to filter<br>Ambient temperature more likely to support bioburden<br>High temperatures may identify any potential compatibility issues |
| | Duration | Longer | Extended durations may increase possibility of bacterial penetration. Consideration should be given to the total filter/product contact time (e.g., account for time needed for any potential product-wet integrity-testing prior to filtration) |
| | Pressure | Higher | Higher pressures may contribute to bacterial penetration under specific conditions (e.g., higher-risk fluids) |

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| Description | Parameters | Worst-Case | Rationale |
|---|---|---|---|
| **Process Conditions** | Flow rate | Higher | Higher flow rates may contribute to bacterial penetration under specific conditions (e.g., higher risk fluids) |
| | Throughput | Higher | Increased volume throughput may increase potential prefiltration bioburden to the filter |
| | Intermittency | Case Specific | Consider if pressure spikes following an “off” cycle could impact potential filter performance; exact simulation of “on/off’ cycles may not be possible in scale-down test |

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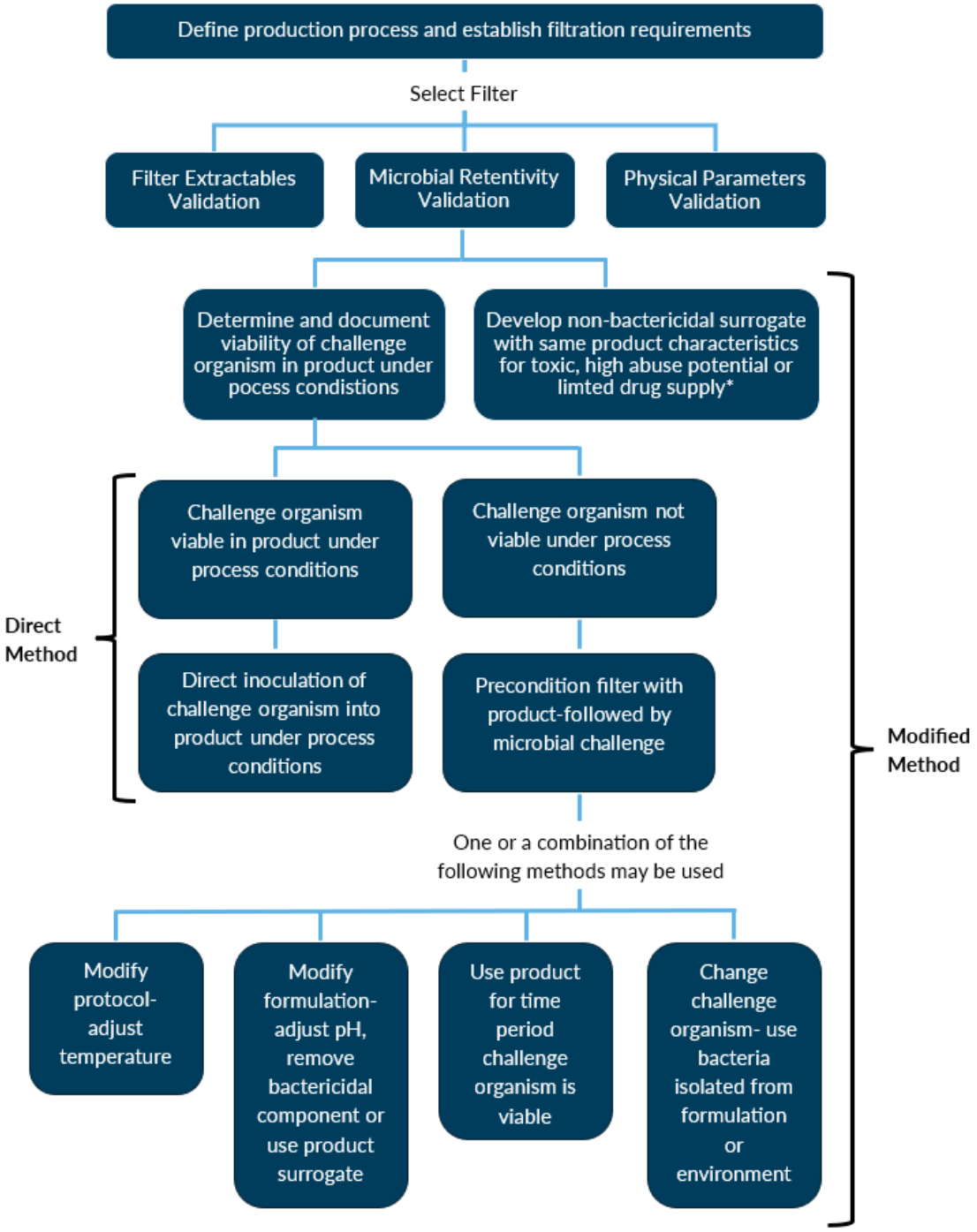
### 6.6.7 Factors to Consider for Bacterial Retention Study Design

If questions arise during protocol development regarding the acceptability of a testing methodology, it may be advisable to contact the appropriate regulatory agency for guidance. **Figure 6.6.7-1** reflects key steps to be considered when selecting the appropriate validation strategy for a specific filter and product/process combination.




***Note:** Concurrence of the appropriate regulatory agency should be sought prior to using this methodology.

**Figure 6.6.7-1 Sterilizing Filtration Process Validation Strategy**

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Irrespective of the test design used, it is important to include a 0.4/0.45 µm positive control filter in parallel with the test filters to assure organism size, monodispersion and viability. Other modifications for the 0.45 µm control filter may be appropriate if fully justified and still demonstrate the challenge organism is fit for purpose.

### 6.6.7.1 Non-bactericidal Processes and Fluids

Direct inoculation of the product with the challenge microorganism is the preferred method of validating microbial retention of a sterilizing-grade filter. This is possible with products and process fluids that have demonstrated no bactericidal effects from the product or processing conditions. For these processes, the product should be directly inoculated with the challenge organism at sufficient concentration and under simulated processing conditions, including time, pressure, flow rate, volume throughput and other critical variables (e.g., temperature) as indicated in **Table 6.6.6-1**. Dilution from bacterial inoculation should be minimized to prevent altering the product any more than necessary.

### 6.6.7.2 Bacteriostatic/Bactericidal/Nondispersive Challenge Fluids

Performing bacterial retention studies on bactericidal products makes it more difficult to answer questions relating to validation, such as What effect does the product have on the filter and what effect does the product have on flora within the product? Bacterial retention studies performed on a bactericidal formulation, or under challenge conditions adverse to microbial viability (e.g., elevated temperature), may not produce valid results.

To evaluate the potential effect of the product/process on the filter, the filter may be preconditioned with the product under simulated processing conditions, including flow rate, pressure, temperature, volume throughput and time. This preconditioning may be performed by recirculating the product through the test filter in a closed-loop system or by a single pass (which could include a static soak to meet required contact time) through the test filter, followed by the bacterial challenge. **Section 6.6.7.3, Section 6.6.7.4,** and **Section 6.6.7.5** discuss examples of test-method modifications that may be used to accommodate this type of bacterial retention study. Other modification methods may be equally appropriate, however, if justified.

### 6.6.7.3 Reduced Exposure Time

In some instances, the challenge organism may not survive in the product for the total processing time. The product should be inoculated directly with the challenge organism at sufficient concentration near the end of the preconditioning phase of the challenge (see **Figure 6.6.7-1**). The duration of the bacterial retention study should be supported by the viability data.

An alternative is to expose the challenge microorganisms in the challenge fluid under static conditions. After exposing the challenge bacteria in the product at process temperature and preconditioning the filter by recirculating the product (without the bacteria) under simulated processing conditions, the filter could be challenged under worst-case processing conditions (e.g., differential pressure and flow rate). The challenge duration should be minimized to ensure the challenge microorganisms remain viable during the actual exposure, challenge, and recovery periods.

### 6.6.7.4 Modify Test Method Parameters

Test-method parameter modification is useful for challenging bactericidal products, since it can involve changing only one process variable, such as temperature. Using this approach maintains the product/challenge organism interaction. This approach does not account for all possible process/product

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interactions but should allow the use of the standard challenge organism. After preconditioning the filter with the product under actual process conditions, the bactericidal component of the process (e.g., temperature) is modified and the bacterial retention study performed. The product should be inoculated directly with the bacterial retention study microorganism at a sufficient concentration (see **Figure 6.6.7-1**).

### 6.6.7.5 Modifying Test Product Formulation

Another modified method is to remove the bactericidal component from the product for the bacterial retention study following preconditioning of the filter with the actual product. This may be as simple as adjusting the pH to a nonbactericidal range, removing or diluting the bactericidal component, or using a surrogate fluid. Whatever approach is used should be supported by a technical rationale.

The modified formulation should be inoculated directly with the challenge organism at a sufficient concentration under simulated product processing conditions (see **Figure 6.6.7-1**). The bactericidal agent should be removed to a level that eliminates its interference with the challenge microorganism.

Another option is to precondition the test and control filters with the bactericidal process fluid under simulated worst-case processing conditions. After removing the bactericidal fluid from the test rig, the filters can then be challenged with the test organism suspended in surrogate fluid that has been justified to resemble the process fluid (e.g., chemical or physical properties). Other test modifications may also be acceptable if justified.

### 6.6.8 Filtrate Sampling

Analysis of the entire challenge effluent is necessary to determine the ability of a sterilizing-grade filter to retain the bacterial challenge. This can be done either by direct passage through an appropriate analytical membrane installed downstream of the test filter (whereby any penetrating bacteria would be captured on the analytical membrane) or by filtrate collection in a sterile vessel and subsequent filtration through analytical membrane(s). The analytical membranes should have been qualified as fit for purpose and will usually have a maximum rating of 0.4 µm or 0.45 µm. Selection of the size and porosity should be based on the application conditions and should not prevent achieving the desired differential pressure across the test filter. Sampling a portion of the filtrate is insufficient to validate sterilizing filtration, because a small number of cells may have penetrated the filter and remain undetected in the portion of the filtrate not sampled and analyzed.

### 6.6.9 Results Interpretation

Acceptance criteria for sterilizing-grade filter performance are met when no passage of the bacterial challenge is detected in three test filters with a valid positive control (i.e., bacterial recovery through the 0.45 µm-rated control filter). If passage is detected in one filter, and no assignable cause can be determined, retesting may be conducted (e.g., three filters from the failed membrane lot) following an investigation and risk assessment to determine a potential root cause. If bacterial penetration occurs in any of the triplicate tests, then the filter cannot be considered validated under the specified conditions. If an assignable cause for the failure can be determined (e.g., damage to the filter or breach in aseptic technique), the suspect lot of filters should also be retested for confirmation of the proposed root cause for the initial failure. If the filter fouls prior to achieving the target volume throughput, the bacterial retention study should be terminated and the resultant total challenge determined. An investigation into the cause of fouling should be performed and

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documented, and a new bacterial retention study design should be proposed to ensure volume throughput is met based on filterability test results.

## 6.7 Considerations for Sterilizing-Grade Filters Configured in Series

When two final sterilizing-grade filters are configured in series, it is necessary to clarify whether the setup is redundant filtration or double filtration, sometimes referred to as additive filtration (see **Section 5.5.1** for detailed descriptions of various filter train designs). When redundant filtration is used (i.e., only one filter is required to meet sterility and only one filter must meet post-use integrity-test specifications), the process-specific bacterial retention study should be based on only one filter. Double filtration means both filters are required to achieve sterility (usually determined from process-specific bacterial retention study), and both filters must therefore meet post-use integrity test specifications.

Regardless of whether the setup is redundant filtration or double filtration, extractables should be considered for both filters.

## 6.8 Chemical Compatibility of Filter

While compatibility of the filter is assessed during initial filter-selection by the end user, compatibility testing may also be performed as part of the final sterilizing-grade filter validation testing. Chemical compatibility testing should encompass the entire device and depends on the fluid, filtration temperature and contact time.

A summary of worst-case conditions for conducting compatibility testing is shown in **Table 6.8-1**.

**Table 6.8-1 Worst-Case Considerations for Compatibility Testing**

| Description | Worst-Case | Rationale |
|---|---|---|
| Temperature | Higher | Higher temperatures increase potential for chemical reactions |
| Duration | Higher | Longer durations may increase potential for chemical reactions |
| Properties of drug product | Consider pH extremes, solvent and product components | Based on risk assessment |
| Sterilization | Maximum sterilization conditions (e.g., maximum temperature, number of cycles) | Worst-case sterilization conditions may impart more stress on the filter |

An assessment of chemical compatibility could include bacterial retention study data, post-exposure integrity-testing, tensile strength, nonvolatile residue (NVR), extractables, particulates, flow rate, optical or scanning electron microscopy (SEM), burst pressure, and membrane/O-ring thickness ***(23)***.

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## 6.9 Revalidation Considerations

Once a filter is validated for use in a given application, further validation may be required if changes to the filter or manufacturing process are made. Some changes that may require revalidation include, but are not limited to:

- Change in membrane type or filter materials of construction.
- Increase in process parameters (e.g., volume to be filtered through a given filter area, flux, pressure, duration)
- Change in filter configuration (e.g., cartridge to encapsulated device). While the process-specific bacterial retention study data will likely not be impacted, consideration should be given to a reassessment of extractables/leachables and compatibility.
- Product formulation changes (e.g., change in product concentration, pH)
- Change in sterilization method (revalidation of the extractables/leachables assessment and filter-integrity post-sterilization should be considered based on risk assessment with appropriate justification)
- Temperature of filtration.
- Use of a filter with the same membrane type but not from the same supplier.

A well-structured risk assessment should be performed by process and filtration experts to evaluate the potential impact of these changes. In line with internal policy for filter management, the filter end user should approve all changes that potentially impact the current good manufacturing processes (GMP) compliance of the system.

## 6.10 Miscellaneous

With the introduction of newer manufacturing modalities, such as advanced therapy medicinal products (ATMPs), additional factors may need to be considered for the purposes of filter validation. For example, small batch sizes, short shelf life, and physical properties (e.g., shear sensitivity) may require careful planning and potentially modified test procedures to achieve the goals of filter validation while meeting regulatory expectations.

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# 7.0 Integrity Testing

Bacterial retention studies are used for validating a filter membrane but are not feasible for routine confirmation of the retentive capabilities of each filter used in the manufacture of sterile pharmaceuticals. Such challenge tests can clearly not be performed prior to the use of the filter in the production setting, as it involves deliberate introduction of contaminating microorganisms to the filter (a "destructive" test). It is also not feasible post-use, as it requires advanced microbiology equipment and multiday incubation periods. Hence the need for a nondestructive, physical-integrity test method that can be correlated with bacterial retention (i.e., can be used as a surrogate method for direct bacterial retention). The main objective of a nondestructive physical integrity test is to determine the presence of defects that may compromise a filter's retention capability without rendering the filter useless. Additionally, the integrity test establishes the similarity of the test filter to validated bacterial retention-challenged filters under process-related conditions.

For integrity tests on hydrophilic membranes, gas flow properties of wetted filter membranes are evaluated over a range of pressures. This can be demonstrated by completely wetting the entire filter membrane and introducing gas to the upstream side of the membrane at a low pressure. Capillary forces within the filter material keep the liquid from being expelled from the pores, and the wetted membrane functions as a barrier to convective gas flow. As this pressure is applied on the upstream side of the filter, a small amount of the gas dissolves into the wetting liquid, diffuses across the wetted membrane, and is released on the downstream side when it is maintained at atmospheric pressure (**Figure 7.0-1**). The diffusion rate increases as the pressure on the upstream side is increased. Once a critical pressure is achieved ("the bubble point"), the effect of capillary forces keeping the liquid in the pores will be overcome by sufficient pressure, pushing the wetting liquid out through the membrane pores. This happens first for the largest pores, where capillary forces are the weakest (**Figure 7.0-2**). If the rate of gas flow that passes to the downstream is measured at increasing pressure values, the membrane characterization curve (**Figure 7.0-3**) can be obtained for the given membrane filter.

Characterization of these dynamics (both the diffusive flow rate at pressures below the bubble point, as well as the exact bubble point pressure) can indicate integrity of the filter in a nondestructive manner, resulting in a method that can be performed on every filter used in production. Diffusive flow theory is discussed further in **Section 10.1**. **Figure 7.0-1** illustrates gas diffusion through the wetted membrane pores at pressures where the wetting fluid is held in the pores by capillary forces. **Figure 7.0-2** illustrates gas flow through the membrane at a pressure exceeding the bubble point where the wetting fluid is expelled from the largest pores.

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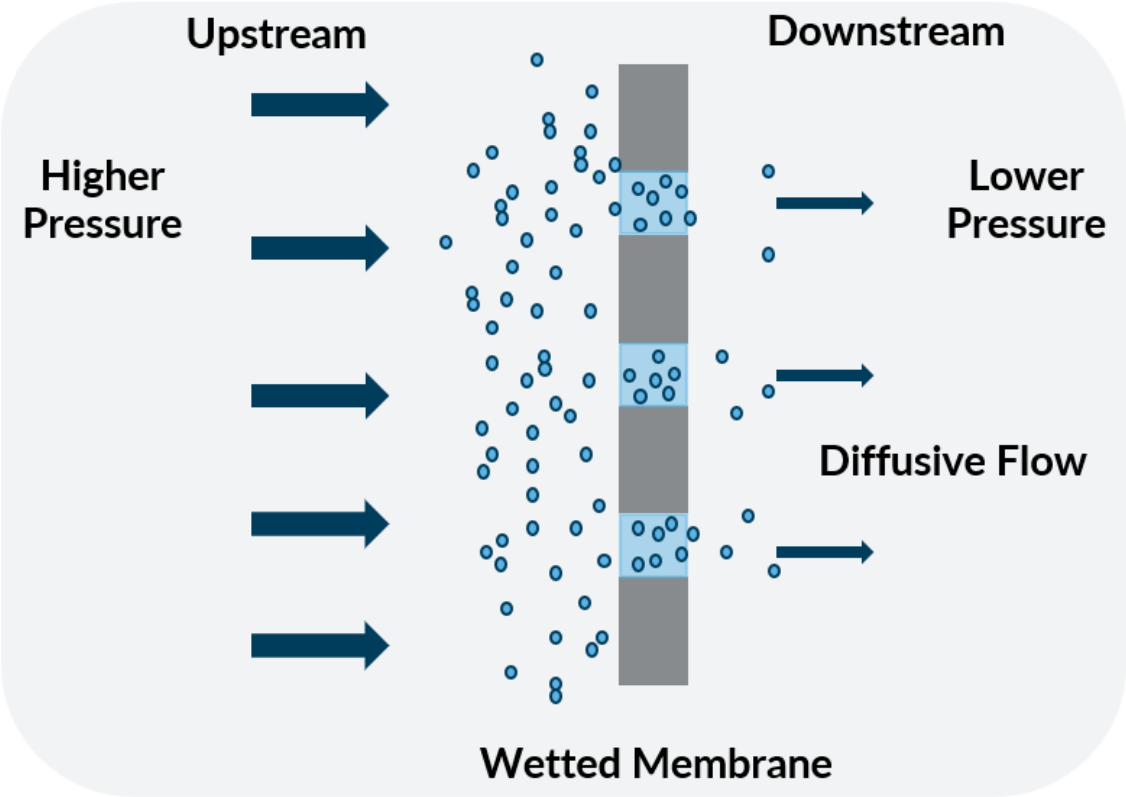


**Figure 7.0-1 Gas Diffusion Through a Wetted Membrane**

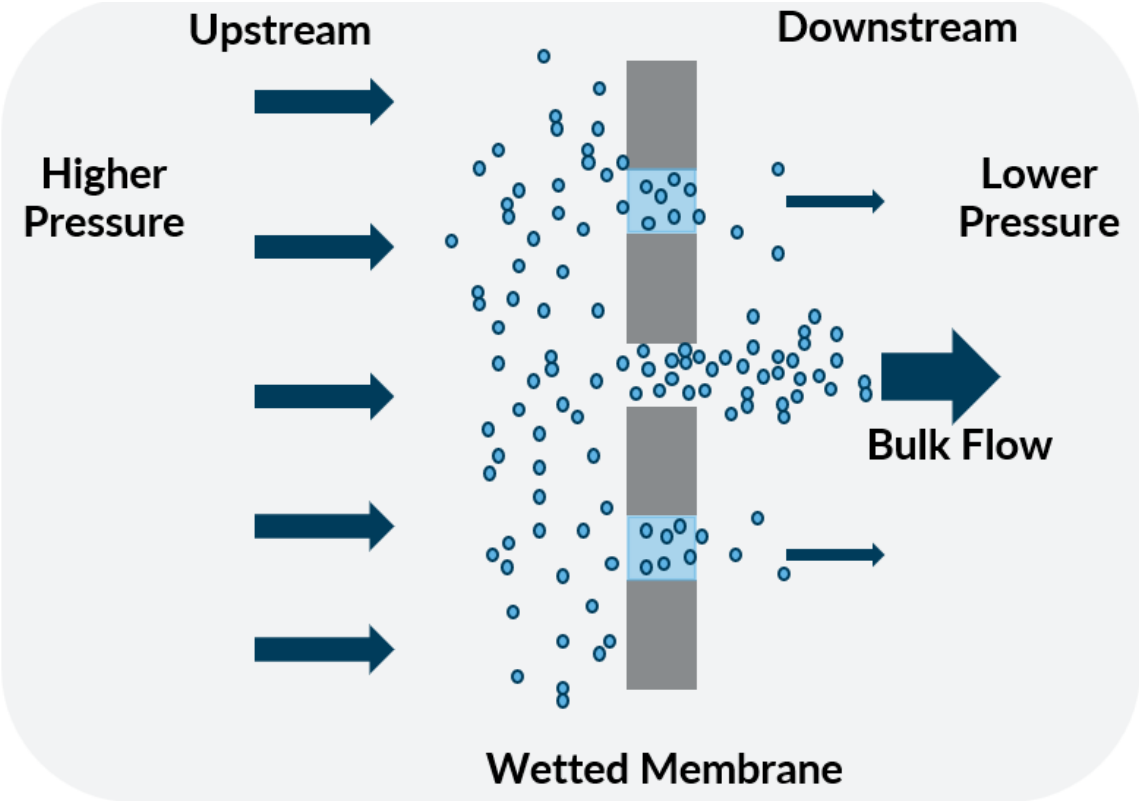


**Figure 7.0-2 Bubble Point Through a Wetted Membrane**

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**Figure 7.0-3** illustrates the relationship between measured air flow downstream of a wetted filter membrane. Three characteristic portions of the curves act as the basis for membrane-filter integrity-testing:

1. The linear portion on the low end of the pressure axis shows diffusive gas flow through the liquid held in the pores of the membrane.
2. As the pressure is increased, there is a characteristic bend in the curve followed by another linear portion. This bend indicates the transition between diffusive gas flow and bulk, or viscous, flow.
3. For an integral filter, bulk gas flow occurs after the bubble point of the largest pores has been exceeded. Before reaching this point, the majority of gas flow is due to free-flowing gas through open pores; a minor portion of the flow results from diffusion through the pores of the membrane that are still wetted.

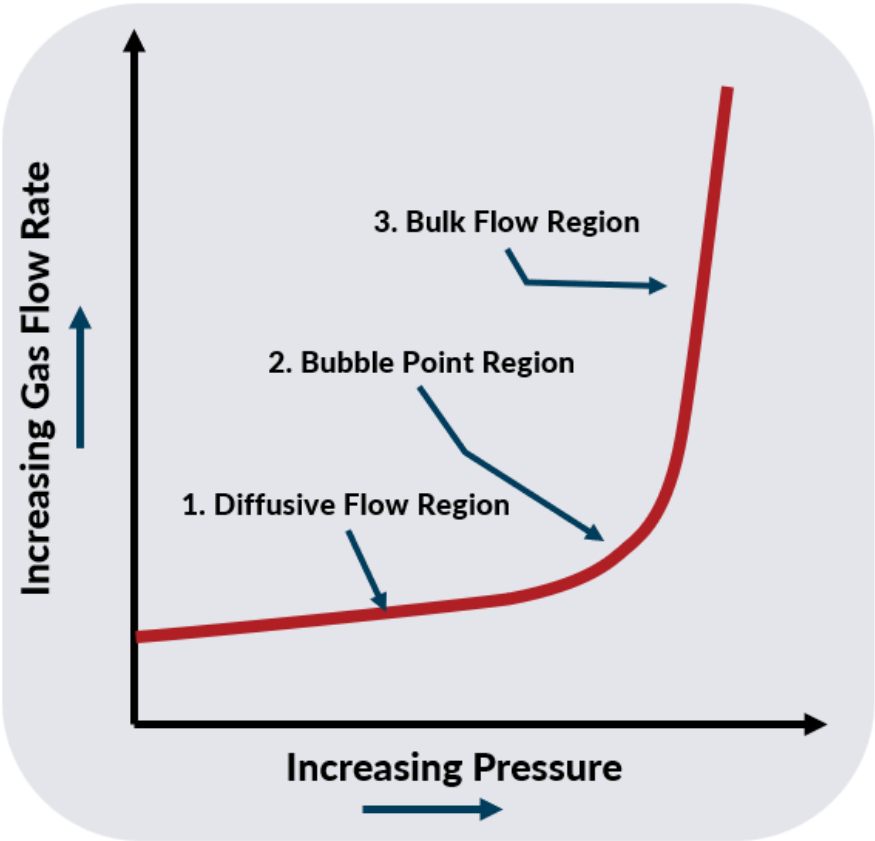


**Figure 7.0-3 Characteristic Curve of the Membrane**

## 7.1 Relationship Between Integrity Test Results and Bacterial Retention

During membrane development by the filter manufacturer, retentive capability can be assessed by challenging successively "tighter" samples of the specific membrane under standardized test conditions and analyzing the bacterial passage results. Typically, the retention pattern observed will allow identification of physical integrity-test specifications using standard wetting fluids—acceptance criteria for the measured parameters within which no bacterial penetration will occur under standardized testing conditions ***(6)***. The integrity-test value established by this exercise is linked to the sterility of the filtrate. Such establishment of integrity-test value specifications should incorporate a safety margin to account for natural variability in measurement and membrane manufacture (e.g., by applying appropriate statistics with confidence intervals).

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**Figure 7.1-1** illustrates a typical relationship between the gas flow profile values and filtrate sterility from a series of five filters with decreasing maximum pore size (from left to right).

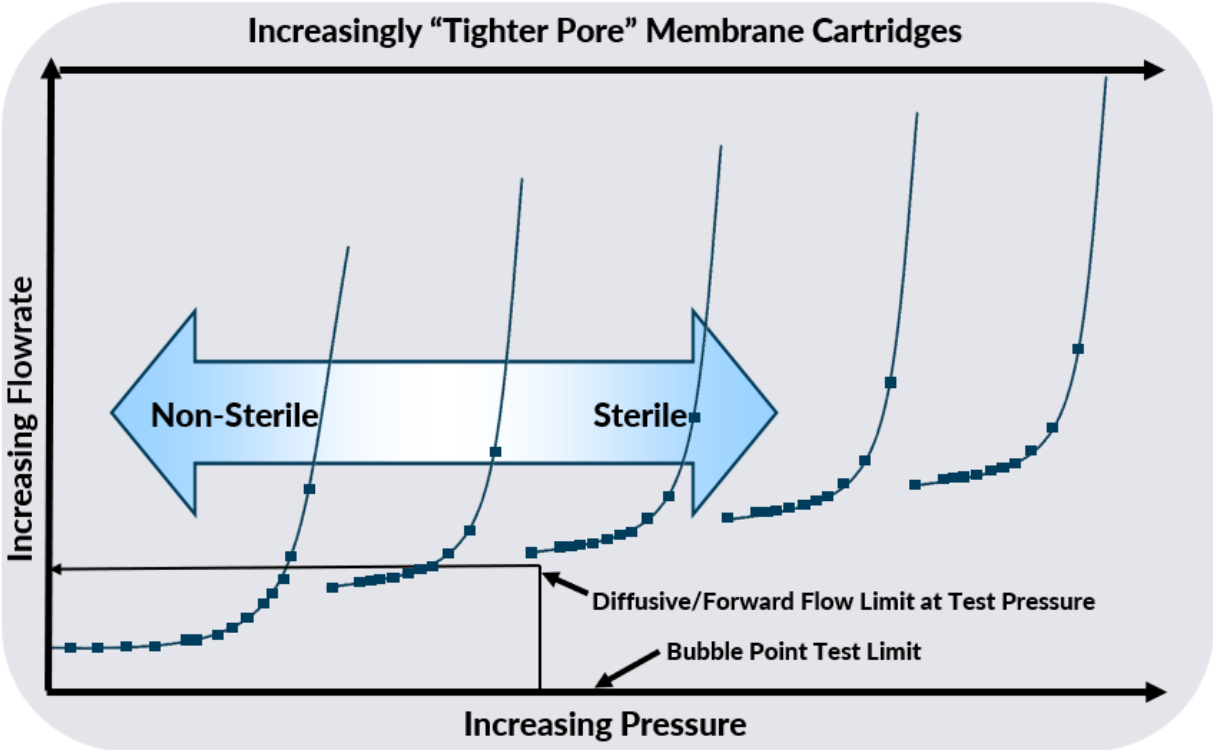


**Figure 7.1-1 Gas Flow Profile Values and Filtrate Sterility**

The bubble point test relates to the effective diameter of the largest pores present in the membrane, and these pores, along with membrane thickness and pore tortuosity, directly influence the retention properties of the membrane. The resolution of the bubble point test diminishes as a function of increasing filter area, since diffusive gas flow below the bubble point tends to obscure it ***(41-43)***.

While diffusion does not bear a direct relationship to pore size, the diffusive flow test provides a quantitative measurement in which maximum flow limit is established by the filter manufacturer at a designated test pressure lower than the minimum bubble point value.

During membrane development and initial integrity-test qualification by the filter manufacturer, testing flow rate at multiple pressure values enables plotting of the gas flow curve from diffusion at low pressures through the bubble point region at elevated pressures. These tests combine the advantages of the bubble-point and single-point diffusive-flow integrity tests in characterizing the pore size distribution. Methods for mathematically characterizing the diffusive-flow to bulk-flow transition region and correlation of the value generated to bacterial titer reduction may be found in reference literature.

### 7.1.1 Test Method Selection

The method for integrity-testing of production filters by the filter user should be chosen to provide reliable results based on the nature of the filter, product, and processing conditions. Bubble point, multiple-point and single-point diffusive-flow and pressure-hold tests can be used, recognizing that each has strengths and limitations that must be evaluated in terms of the particular circumstances of the test. These test methods can be used for both hydrophilic and hydrophobic membrane filters and can be performed manually or

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preferentially with automated integrity-test instruments. **Section 10.2** contains further discussion of filter integrity-test methods.

The flow rate of the test gas that passes through the wetting fluid in the pores of the filter media via diffusive flow is much lower than the flow rate of bulk gas that passes through the filter media when the wetting fluid is displaced from the larger pores of the filter media (i.e., what occurs at and above the bubble point). The rate at which test gas passes through the filter media via diffusive flow is directly related to the surface area of the filter media (i.e., as the filter surface area increases, the flow rate due to diffusive flow increases).

Integrity-test instruments have a lower limit at which they can reliably determine the flow rate of the test gas passing through the filter media. For this reason, there is typically a minimum value for the surface area of the filter being integrity-tested to perform a diffusive-flow integrity test. Below this minimum surface area, the automated integrity tester may not be capable of determining the diffusive flow at the specific test pressure (i.e., <= 80% of bubble point). In contrast, when using the bubble point method, the algorithm is looking at the change in the flow rate instead of the absolute flow rate as the differential pressure is increased (specifically, the pressure at which bulk flow starts). For this reason, the bubble point integrity-test method is typically used instead of the diffusive-flow integrity-test method to perform an integrity test on a filter with a small surface area.

Above the minimum surface-area threshold for testing via the diffusive-flow integrity-test method, either the diffusive-flow or the bubble point test method may be used for a range of filter surface areas. As the surface area of the filter media being tested increases significantly (e.g., in the case where multiple round housings are used), there comes a point where the diffusive flow of gas passing through the wetted membrane is so high that it will be difficult for the integrity test to distinguish the bubble point) from the high rate of "background" diffusive flow through wetted pores. For this reason, the diffusive-flow method is typically selected for the largest filter sizes.

The exact cutoffs for small, medium, and large surface area depend upon the attributes of the filter media being tested, the wetting fluid used, and the capability of the automated integrity tester being used, but general guidance is provided in **Table 7.1.1-1**.

**Table 7.1.1-1 Suggested Integrity Test Methods Based on Filtration Surface Area**

| Surface Area of Filter Media being Integrity Tested* | Suggested Integrity Test Method Based on Effective Filtration Area |
|---|---|
| < 0.03 $m^2$ | Bubble Point |
| 0.03 – 1.8 $m^2$ | Bubble Point or Diffusive Flow |
| > 1.8 $m^2$ | Combination of Diffusive Flow and Bubble Point |
| *These values are approximate values only. Consult Filter Manufacturer for specific guidance. | |

One approach to diffusive-flow testing of large filter systems with multiple cartridges is to derive an assembly-specific value that is lower than the linear multiplied-limit of each cartridge. A technical evaluation should be conducted to justify this approach. If the assembly's diffusive air flow exceeds the

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corrected limit, an investigation would be required (e.g., individual tests on each cartridge, and confirmation of the seals with the housing).

When performing tests on filter capsules with associated SUS, the high pressures required for bubble point testing could exceed the pressure ratings of such equipment. For this reason, diffusive-flow tests may be preferred as they typically require lower pressures than bubble point-test methods. Or, in case the bubble point method is required, the SUS is designed to be compatible with a higher working-pressure range.

A modification to the diffusive flow method is a pressure-hold/decay method, where pressure is allowed to drop as diffusive flow across the wetted membrane occurs, instead of being held constant (see **Appendix B** for more information).

### 7.1.2 Filtrate Sterility Assurance

Integrity-testing of sterilizing-grade production filters is a fundamental element of sterility assurance for the most critical applications that require a sterile claim ***(9, 11)***. However, integrity-testing alone is insufficient to ensure the sterility of the filtrate. Other control elements should also be in place, such as:

- Filter manufacturer's production controls and quality assurance systems should be in place to ensure the quality and uniformity of the filter membranes and fabricated filters.
- Filter user should conduct bacterial retention validation studies to demonstrate that product, processing conditions, and sterilizing-grade filter can effectively sterilize the fluid.
- Filter user should ensure appropriate product and process controls (e.g., system design, operating parameters, bioburden control, process monitoring) are in place so that the filtration operates within the validated parameters defined by the bacterial retention study.

With the appropriate controls in place, the filter integrity test is only one part of the overall sterility-assurance strategy. It is incumbent on the filter user to ensure that all elements are, and continue to remain, in place.

## 7.2 Automated Integrity-Test Instruments

The standard for modern filter integrity-testing utilizes automated integrity-test units, which provide the most reproducible and sensitive methods. Automated integrity-testers employ highly sensitive pressure sensors and/or gas flow meters, in combination with the careful regulation of pressure, to test filters in a highly repeatable manner.

Historically, filter integrity-testing had been performed manually via a combination of manual manipulation of pressure regulators, graduated cylinders, and/or visual detection of increasing gas flow rate via a stream of bubbles. Some manual integrity-test methods require downstream manipulations that could compromise the sterility of the system if performed before use. Automated integrity-test instruments perform the integrity test from the upstream (nonsterile) side of the filters, minimizing the risk for downstream contamination. Manual integrity-testing should now only be performed on production filters in special and justified cases, due to the ready availability of automated integrity-test instruments.

Automatic integrity-test equipment, both hardware and software, should be qualified, as described in **Section 7.3**. Most automated integrity-test instruments function by measuring pressure only upstream of the filter membrane, and *assuming* downstream pressure is atmospheric. Care must be taken if the downstream side of the membrane is not in equilibrium with the atmosphere.

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## 7.3 Qualification of Integrity Test Devices

Qualification of integrity-test devices is similar to qualifying other process test-equipment. Qualification of an integrity-test device begins with the development of the device and is usually conducted by the device manufacturer. Design qualification and device development documentation are generally generated by the device manufacturer. These documents may be included in the filter user's qualification documentation package. Equipment qualification should cover the key functions of the device; however, it may not cover all possible combinations of functions and configurations. Risk assessment, therefore, should include key functions such as, but not limited to:

- Test procedures as a main function of the device
- Detection of error conditions and messages
- Compliance with data integrity requirements and good documentation practices
- Qualification of the measurement itself (accuracy and limits)

Typical qualification activities may include:

- Computer software evaluation–test parameters, test methods, programming the unit, and tests (e.g., ease of use, test options, networking capability, operating systems)
- Unit sensitivity evaluation
- Unit startup
- Unit calibration
- Performing the tests
- Integrity-test performance evaluation (bubble point, diffusive flow, pressure hold)
- Test other functions (volume determination, failure modes, ejecting invalid entries)
- Test printout evaluation
- Password protection in combination with appropriate user roles and permission levels to avoid modification of a test program by an unauthorized user.
- Peripheral function evaluation (date/time clock, memory, cleaning)

Integrity-test devices that measure pressure and/or gas flow should be calibrated using relevant calibration standards and should generate records according to applicable regulatory expectations.

Automated integrity-testers often incorporate sensitivity settings and test algorithms to detect atypical test conditions and to reliably test different filter sizes. These sensitivity settings are of importance and require careful observation to assure the right sensitivity for the particular filter-size is chosen, otherwise the integrity test may lack accuracy and validity. The sensitivity parameter establishment can be performed or confirmed during process qualification (PQ). For thorough qualification strategy, end users should consider confirming not only that the unit can successfully return a "pass" integrity result for integral filters but can also reliably detect broken filters or incorrect test setups.

## 7.4 Product Wet Integrity Tests

The default "reference" wetting fluid for physical integrity tests of hydrophilic filters, as validated by membrane manufacturers, is typically water. However, for testing production filters, drug product or buffers may be the most appropriate wetting fluid (e.g., after use, when the filter is already wetted with this fluid). A new integrity-test specification, customized for the unique wetting fluid, can be determined during filter validation by executing a protocol in which the reference fluid specification is "converted" to an appropriate specification for the new wetting fluid. This specification is then, by extension, also correlated to the bacterial retention capabilities of the membrane.

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Differences between product-wetted integrity test values and reference fluid-wetted integrity test values are due to differences in test gas solubility, the diffusion constant, and the surface tension of the wetting fluid.

The scale-down studies are used to determine the correction factor between the water-wet integrity-test specification and the required product-wet test limits.

### 7.4.1 Product-Wetted Bubble Point Tests

Tests for determination of product-wet bubble point limits should be conducted, first with water followed by product-wetting and then by the product-wet test. The technical rationale for establishment of product-wet bubble point values should be documented. Bubble point values are affected by the surface tension and temperature of the wetting fluid. **Section 7.4.1.1** describes one method used to establish bubble point specifications using filter discs. Other methods may be appropriate if scientifically justified (see **Section 10.2** for further discussion of bubble point tests).

#### 7.4.1.1 Bubble Point Ratio Approach

For determining product-specific bubble point specifications, testing can generally be conducted as follows:

1. Select multiple samples of filter discs for testing—generally at least three membrane lots. Since the bubble point refers to specific values relating to the membrane's pore size and the wetting-liquid surface tension, filter discs can be used.
2. Install the membrane discs in the holder and rinse them with the reference wetting liquid (e.g., water, alcohol).
3. Perform the bubble point test on each disc.
4. Either completely dry the filter discs or rinse the membranes with an appropriate amount of the product in question to ensure complete removal of the wetting fluid used in the first integrity test.
5. Perform the bubble point test with the product. These values are used to calculate the product-wet bubble point limit.

The bubble point limit for product can be calculated using the formula given in **Equation 1**. This example uses water as the reference fluid, which correlates the product-wetted-filter bubble point test data to the minimum water-wetted bubble point value given by the filter manufacturer.

**Equation 1:**

$$PBP_{min} = WBP_{min}\left(\frac{PBP_{avg}}{WBP_{avg}}\right)$$

**Where:**

$PBP_{min}$ = Product-wetted bubble point, minimum (minimum validated product-wetted bubble point)

$WBP_{min}$ = Water-wetted bubble point, minimum (minimum water-wetted bubble point specified by the filter supplier)

$PBP_{avg}$ = Average product-wetted bubble point of the test samples recorded in the test

$WBP_{avg}$ = Average water-wetted bubble point of the test samples recorded in the test

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### 7.4.2 Product-Wetted Diffusive Flow Test

The determination of product-wetted test specifications for diffusive flow is a two-step process, which requires first, identifying the bubble point ratio as described in **Equation 1**.

#### Step 1: Determination of Test Pressure

The test pressure for the product-wetted diffusive flow test is typically calculated using the following formula and the determined product wet bubble point values:

**Equation 2**

$$TP_{PW} = MTP_{WW}\left(\frac{PBP_{avg}}{WBP_{avg}}\right)$$

**Where:**

$TP_{PW}$ = Test pressure, product-wetted (varies with surface tension)

$MTP_{WW}$ = Manufacturer test pressure, water-wetted (or other test solvent) (see **Section 10.2**)

$PBP_{avg}$ = Average product-wetted bubble point of the test samples recorded in the test

$WBP_{avg}$ = Average water-wetted bubble point of the test samples recorded in the test

#### Step 2: Determination of the Product-Wetted Diffusive Flow Limit

The product-wetted diffusive-flow limit is then typically determined as follows:

1. Conduct replicate water-wetted diffusive-flow tests on the filter assembly at the test pressure specified by the filter manufacturer (small-scale devices may be used as long as an accurate determination of flow is possible).
2. Dry the previously water-wet tested filter assembly and conduct several replicates of the product-wetted diffusive-flow test at the appropriate test pressure determined in **Step 1, Equation 2** above.
3. Calculate the diffusive-flow test limit using the formula expressed in **Equation 3**:

**Equation 3**

$$DFL_{PW} = DFL_{WW}\left(\frac{DF_{PW}}{DF_{WW}}\right)$$

**Where:**

$DFL_{PW}$ = Diffusive flow limit, product-wetted

$DFL_{WW}$ = Diffusive-flow test limit specified by the supplier, water-wetted

$DF_{PW}$ = Diffusive flow, product-wetted

$DF_{WW}$ = Diffusive flow, water-wetted

Diffusion values are affected by the solubility of the test gas in the wetting fluid and its diffusion constant through the wetting fluid, therefore acceptance criteria are specific to a particular test gas being used.

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## 7.5 When a Sterilizing-Grade Filter Should be Integrity-Tested

Where the claimed purpose of the filter is to sterilize, post-use integrity-testing is mandatory. This confirms that the filter remained integral throughout its use and was capable of retaining microorganisms. If truly nonintegral filters are detected post-use, a thorough investigation to determine the root cause is required. Guidance on best practice is to leverage the supplier's or manufacturer's expertise. In most cases, it also results in discarding a drug batch (or reprocessing, where validated) to protect patient safety from exposure to a nonsterile product.

For filters marketed for critical sterilizing filtration, each filter element should also be integrity-tested by the filter manufacturer prior to release.

Integrity-testing can also be performed by the end user prior to use, either before the filter itself is sterilized (when sterilized by the end user) or after sterilization (**Figure 7.5-1**). This allows early detection of nonintegral filters or improperly installed filters prior to execution of the filtration process. Whether to test filters in-line or off-line will depend on actual process requirements, but the test must include the cartridge already installed in the filter-housing (if applicable) so the seal between the filter element and the housing is also confirmed. Regulatory expectations regarding pre- and post-filtration integrity-testing may differ regionally ***(9, 11)***. In general, the objectives of pre- and post-filtration integrity-testing vary, and the need for each should be informed by risk assessment.

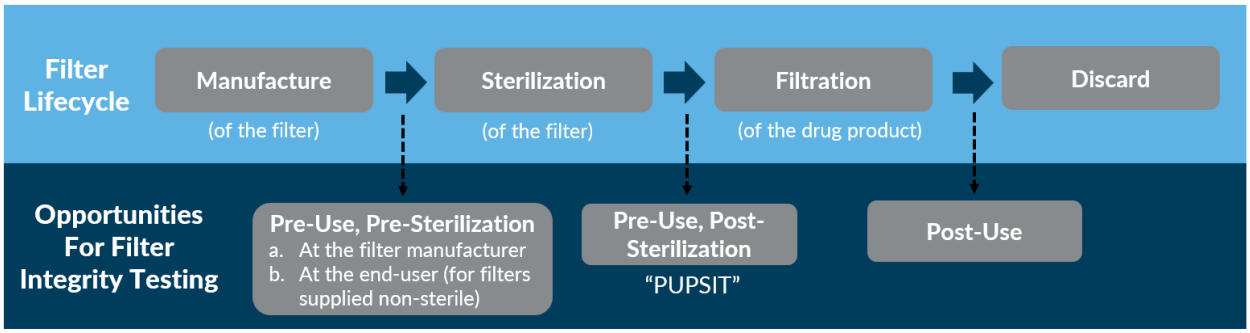


**Figure 7.5-1 Filter Lifecycle–Opportunities for Integrity-Testing**

Pre-use, pre-sterilization integrity tests performed by the end user would be redundant to the integrity test performed by the filter manufacturer to release the sterilizing-grade filter from their production. However, a presterilization test will additionally confirm that an integral filter of the proper pore size has been correctly installed in the housing, and that damage has not occurred during shipping and handling. A post-sterilization integrity test provides the same information as a pre-sterilization test plus confirmation that the filter was not damaged during sterilization. Steps must be taken to ensure that the downstream side of the system remains sterile when performing PUPSIT.

### 7.5.1 Pre-Use Post-Sterilization Integrity Test (PUPSIT) Considerations

The post-use test is designed to detect any filter damage that occurred throughout its lifecycle, up to and including the filtration process itself. However, under extreme circumstances, post-filtration integrity test results may be influenced (i.e., shifted) by high-foulant fluids, which may mask a filter flaw or larger pore structure ***(44-47)***. In this scenario a PUPSIT result could identify filter flaws or filter damage when a post-use integrity may not**.** Recent publications have shown that these conditions are atypical for terminal filtration steps as filter-fouling is typically avoided within these filtration applications. Filterability studies, performed to determine the appropriate filter size and area before any filter design is established, will

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indicate the fluid-fouling propensity and filter flaw-masking risk, and gauge potential preventive measurements against any fouling ***(47)***.

In filtration applications with significant pore-plugging or large-pore structure-blockage potential, it is advisable to know the filter's prefiltration integrity-test value prior to the influences of contact with the full batch of product (or more strictly, the potential of foulants to shift the measured post-use integrity-test results in the direction that might cause false passes.) Any excessive fouling conditions would be also evidenced by significant flux decay or required pressure increase to maintain flow. True post-use integrity-test failures are typically a result of damages due to extreme mechanical or thermal process conditions. Such damages are typically catastrophic, and flaw-masking would not be possible for catastrophic or gross damage to filter membranes, as this damage is not capable of being "masked" by components of the fluid to an extent that allows a post-use test to appear to pass ***(48)***. However, in cases where significant fouling of the filter is possible, it could potentially mask rare marginal membrane flaws, which may not be detected by the post-use integrity test. Excessive filter-fouling could also be detected by special customizations to the post-use test, as the decrease in porosity and blockage of the largest pores would be manifested in a decrease in diffusive flow and in an increase in the bubble point, respectively ***(49)***. Alternative means also exist to detect any fouling that has the potential to modify post-use test value, including flow and differential-pressure monitoring during filtration, and possible modifications to the post-use test to detect unexpectedly low diffusive-flow values or unexpectedly high bubble points. Further, any risk of such flaw-masking (due to product-specific mechanisms of shifting the integrity-test results after fouling) can be evaluated during process design. Decades of experiences of filter suppliers and end-users showed that this type of behavior is scarce and evaluated during filterability tests respectively filter area sizing and process validation.

Any potential risks involved in the sterilizing filtration activity must be thoroughly assessed ***(28, 50, 51)***. While PUPSIT is a common regulatory expectation, there may be process constraints where it is not feasible ***(11)***. In such cases, and in relevant jurisdictions, a thorough risk assessment is required not to justify omission, but to evaluate and manage the risks associated with not performing PUPSIT. This includes assessing the filter sterilization process, supply chain controls, packaging, product characteristics, and pre-filtration steps. The goal of this assessment is to ensure that adequate controls are in place to mitigate the risk of using any non-integral filters.

### 7.5.2 Post-Filtration Integrity-Test Considerations

For operational reasons, the post-filtration integrity test should be accomplished as soon as feasible to avoid any detrimental test influences, for example, filter membrane wetting problems. Product residues should not dry out on the membrane and should be removed, if possible (by sufficiently flushing the filter with an appropriate fluid, post-use), if not testing in product. Filters should not be discarded until the integrity-test results are reviewed. Storage conditions and duration should be qualified to ensure they do not adversely affect the integrity of the filter. In cases of abnormally long delay between the use of the filter and its integrity test, the possibility of false passes should be evaluated.

The post-use test is not only a test of membrane integrity, but of the junction between the filter element and its housing or capsule. The filter element to be tested should therefore remain within the filter-housing to ensure that the filter and its junction with the housing has been integral throughout the filtration process. The post-use integrity test could be performed in-line or off-line within specific integrity-test laboratories within the end user's facility. The post-use test can be performed as a product-wet or water-wet procedure. Confirmation of integrity must be obtained before any product batch is released; therefore, the filter must not be discarded before the test result is obtained. If the filter receives an apparently failing integrity-test

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result, specific activities for retesting or root cause analysis are required and should be documented in the relevant procedure or batch record.

### 7.5.3 Integrity Testing for Serial or Redundant Filtration

Manufacturing processes may be designed with trains of multiple filters, but the purpose of these extra filters varies from process to process (see **Section 5.0**). This purpose determines the need for integrity-testing of each filter. If one filter has been validated to achieve sterilization with a specific product, then the single sterilizing filter must satisfactorily pass integrity-testing after use.

#### 7.5.3.1 Two-Filter Sterilization (Double Filtration)

In double filtration, the filter train is considered to be a sterilizing unit, and all sterilizing filters within the train must satisfactorily pass integrity-testing after use.

#### 7.5.3.2 Pre-Filtration

For prefiltration, requirements for integrity-testing must be determined by the end user in an overall assessment during process design. Depending on the purpose of the filter, integrity-testing (correlated to complete bacterial retention) may not always be necessary.

#### 7.5.3.3 Redundant Filtration

The primary filter (generally the farthest downstream) must be integrity-tested post-use. The secondary (redundant) sterilizing filter may not require post-use integrity-testing unless the primary sterilizing filter fails. In that case, the redundant filter must satisfactorily pass post-use integrity-testing and the reason for the failure of the primary filter should be investigated including confirmation of the sterile boundary. If the sterilizing-filtration process is redundant and registered as having PUPSIT, both filters should undergo PUPSIT for them to be considered comparable.

Performing PUPSIT on both filters in a redundant filtration configuration can be quite complex. Each filter must be tested individually after sufficient wetting; a single test cannot typically be used to test both filters simultaneously. Precautions should be taken to maintain the sterility of the fluid pathway between the two filters, as that region becomes critical for product sterility should the downstream filter fail, and the upstream filter is relied upon for sterility. Controls can include the use of additional sterilizing filters to vent the integrity-test gas from the upstream filter and introduce the test gas for integrity-testing the downstream filter. To test the downstream filter, there must be a valve or other closure between it and the upstream filter. In this type of setup, the integrity-test process-method development should include a risk assessment to ensure that the effects of wetting, or nonintegrity of the product and vent filters, does not interfere with the filter integrity-test result ***(52, 53)***. Further details on these considerations were discussed in PDA's *Points to Consider for Implementation of Pre-Use Post-Sterilization Integrity Testing (PUPSIT*).

## 7.6 Failure Analysis/Troubleshooting

If a sterilizing filter fails an integrity test, it could be damaged, but there may be other causes for the failure that include incorrect assembly (incomplete sealing) and incomplete wetting (see **Section 7.7.1**). Due to the physical mechanisms of filter integrity-testing, "false failures" may occur, and controlled retesting with defined actions may be permitted. When procedures are followed and appropriately justified, an ultimate

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passing result can be treated as a valid result even if initial tests resulted in reported failures. Filter-failure investigations and retest procedures should be documented.

To distinguish between filter damage and possible wetting or other problems, the following verification steps may be taken to determine that the:

- Appropriate integrity test has been selected
- Correct test parameters are being used
- Correct wetting fluid and wetting procedure are being used
- No leaks in the test system
- Filter assembly temperature has remained stable and within specification during testing (e.g., adiabatic effect*) ***(54)***
- Equipment has been properly calibrated ***(55)***
- Test setup is properly assembled and functions properly
- Correct filter has been installed

***Note:** Adiabatic effects can influence the test result if temperature is not held constant. To overcome this, extended stabilization and test times may be required for these systems.

After verifying the above, troubleshooting steps (including re-wetting and re-testing) can be performed to rule out possible false failure causes. Example steps in a troubleshooting procedure include:

- Rewet the filter according to the specifications and repeat the test (see **Figure 7.6-1**, Step 1).

If the filter integrity-test fails again, the following step may be employed:

- Apply more aggressive wetting conditions by increasing the flush volume/time, increasing the pressure differential, and/or applying back pressure (see **Figure 7.6-1**, Step 2).

If the filter-integrity test fails again, the following steps may be employed:

- Perform the integrity test in a lower surface-tension reference fluid to assess filter wettability changes independent of filter integrity (see **Figure 7.6-1**, Step 3).
- If the filter fails using the reference fluid, the filter fails the test.

If the filter passes the integrity test at any point during this failure analysis, it can be considered integral and capable of yielding a sterile effluent. **Figure 7.6-1** provides an example of a decision tree that may be used to evaluate integrity-test failures. The total number of test repetitions allowed should be defined and documented, and all actions should be proceduralized. When the total number of test repetitions allowed is exceeded, investigations should be initiated and documented.

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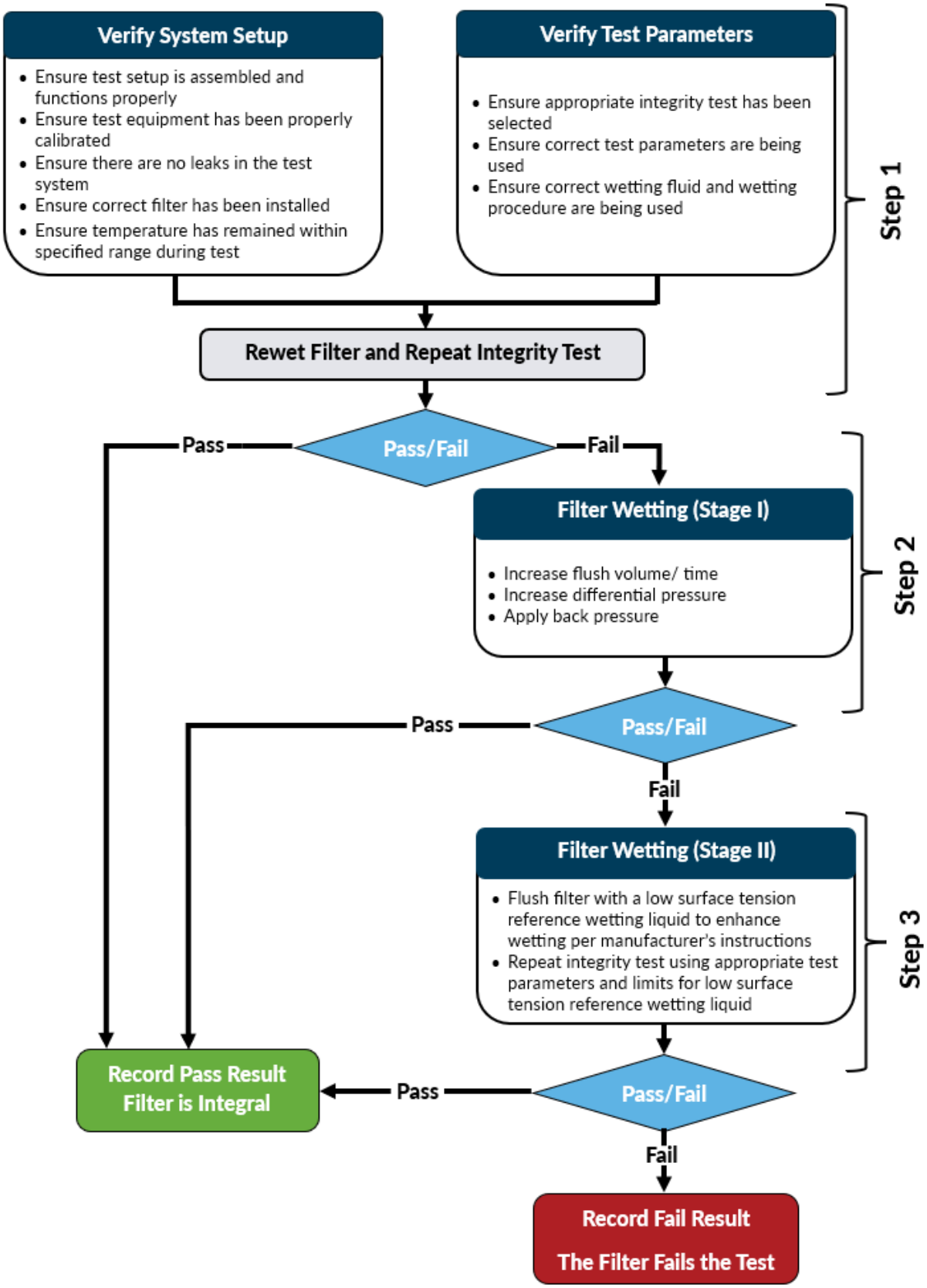


**Figure 7.6-1 Example Integrity-Test Failure Analysis Decision Tree**

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### 7.6.1 Insufficient Wetting Failure Analysis

Filter integrity-test failures are often caused by insufficient wetting of the filter membrane matrix. Incomplete wetting may be due to either inadequate flushing to wet out all pores, adsorption of hydrophobic contaminants, or other formulation components that can change the surface-wetting characteristics of the filter membrane. A change in the wetting characteristics may affect integrity-test performance. For example, leachables from upstream materials may be adsorbed into the filter membrane, impacting its wetting properties and resulting in an integrity-test failure ***(56)***.

To obtain an appropriate integrity-test result, the filter's porous structure must be completely wetted, as the physics of integrity-test measurement rely on either gas flow through the wetting liquid layer or displacement of the wetting fluid from the largest structure. Wetting of the filter membrane can be influenced by:

- **Membrane polymer:** Some polymers are easier to wet than others, depending on the critical surface tension of the membrane material
- **Pore size:** The smaller the pore size, the more difficult the pores are to wet
- **Wetting fluid:** Some wetting fluids may interact unfavorably with the polymeric matrix
- **Product residues or contaminants:** Product residues or contaminants can alter the hydrophilicity of the membrane polymer and repel the wetting fluid or lower the surface tension
- **Pressure conditions:** Manufacturer's pressure recommendations should be followed to completely wet the membrane
- **Temperature conditions:** Temperature influences the surface tension of the wetting fluid

In addition to the above, there may be process- and application-specific factors that could affect integrity tests that are not covered by the manufacturer's description of use. The troubleshooting steps described in **Section 7.6** and the decision tree provided in **Figure 7.6-1** may be useful to evaluate integrity-test failures. Additional troubleshooting information is provided in **Section 10.3**.

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# 8.0 Sterilization of Filters

A key element of a successful sterilizing filtration process is the sterility of the filtration assembly, especially on the filtrate side. Proper sterilization of filters involves applying the right physicochemical conditions, and with careful attention, this process can be highly effective while preserving the integrity of the filter membranes and components. By following the manufacturer's recommendations for sterilization, the risk of thermal, mechanical, or chemical damage to the filters can be minimized. The sterilization process should be appropriately qualified to ensure that there is no variation during the cycle and that it achieves the respective sterility goal and does not cause harm to the filtration device. In instances when the sterilization process is not automated, a routine integrity test of the filter after sterilization may be performed. An integrity test after the sterilization process can confirm that the filter was not damaged during sterilization.

## 8.1 Steam Sterilization

Steam under pressure is widely recognized as one of the most effective sterilization methods, typically carried out in an autoclave or in situ as SIP. Steam sterilization cycles are required to be properly validated and performed to ensure the entire filter and membrane matrix is sterilized. The steam sterilization cycles are often automated and well controlled within the end-user site to assure maintaining the validated state. Validation studies typically include integrity-testing the filters post-sterilization and the use of biological indicators (BI). BI's may be placed in the filter core or embedded into the pleat packs. These studies play a critical role in confirming that the cycle achieves a $10^{-6}$ sterility assurance level and that the filters are integral.

Steam sterilization parameters should be considered carefully when designing a sterilization process for any filter. Refer to the filter manufacturer's literature for additional information regarding temperature and pressure limitations together with the maximum allowable sterilization cycles.

### 8.1.1 Autoclave Sterilization

Most sterilizing filters are qualified for steam sterilization by their manufacturers to at least 121°C and, in some cases, up to 130 – 134°C. Temperatures higher than this may make many of the plastics used for filter construction unstable and may affect the physical integrity of the filter or increase the level of extractables.

Filter assembly preparation for autoclave sterilization is critical. Sterilized filter components should be protected with a suitable microbial barrier to avoid recontamination after sterilization; however, the barrier must allow air to pass from the filter and steam enter the filter during sterilization. It is critical to ensure that filter inlets and outlets are left open to avoid excessive differential pressures during sterilization. Wrapped articles should be covered with only enough wrapping material to protect critical surfaces while allowing free transfer of saturated steam and air through the material. Use of sterilization tape should be minimal. The

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weight of any attached fittings and hoses should be supported to avoid potentially deforming of capsule filter inlet and/or outlet at high temperatures. Hoses should not be kinked or closed to allow the free flow of steam and air. The filter user should refer to the manufacturer's specifications when developing an autoclave sterilization cycle for the filter assembly. Slow exhaust (e.g., liquid) cycles may be considered for filter sterilization to avoid excessive pressure differentials across the filter during the cool-down phase of the cycle. Filters should be removed promptly after the sterilization cycle to avoid potential oxidative damage from heated air in the chamber.

One of the major difficulties encountered in autoclave sterilization is air removal. Air can be effectively removed at the initiation of the sterilization cycle by applying a series of vacuum and steam purge cycles. Large filter assemblies may require adjustments to the sterilization parameters, especially the prevacuum phase, to optimize air removal and steam penetration ***(57)***. During the sterilization cycle, the filter assembly should be oriented to allow condensates to drain. If the filter is integrity-tested immediately post-sterilization, it does not have to be dry.

### 8.1.2 Sterilize-in-Place

The SIP method is limited to filters in housings that can withstand steam pressures of 15–30 psig (1–2 bar). Disposable filter capsules with polypropylene or polyester housings should not be subjected to in-situ steam under pressure, except for high-temperature polymers, such as polyetherimide.

To ensure optimal performance during the sterilization process, it is important to monitor pressure differentials by installing pressure gauges both upstream and downstream of the filter ***(58)***. Adhering to the differential pressure limits set by the filter manufacturer for each specific housing or assembly is critical, especially at elevated temperatures. High differential pressure may occur due to restricted drainage or blocked steam flow but, with proper practices, this can be effectively avoided.

To protect wetted filters, steam should be introduced gently in forward flow at low pressure. This approach allows the filters to heat up gradually, ensuring that any liquid in the membrane matrix is evaporated before steam flow begins. An effective way to facilitate this is by opening the upstream vent and housing drain before the steaming process. This allows steam to efficiently heat the filter, ensuring a smooth transition to sterilization. Once downstream pressure increase is observed, upstream steam pressure can be increased to target the sterilizing pressure by partially closing the vent and drain valves. Alternatively, the system can be designed to introduce steam from both sides of the filter to avoid excessive steam-pressure differentials.

During the cool-down phase of an SIP process, it is critical that air or other suitable gas maintain a positive pressure across the filter. Excessive reverse pressure differential can arise if the upstream side of the filter is cooled more rapidly than the filter core. Gas-assisted cooling is preferred over liquid cooling, because the latter may generate thermal shock and damage the filter ***(58, 59)***.

## 8.2 Irradiation Sterilization

Common irradiation sterilization processes are gamma radiation and X-ray. Gamma and X-ray irradiation sterilization can be validated using a bioburden and dosimetry approach. Follow current International Organization for Standardization (ISO) standards such as *ISO-11137: Sterilization of Health Care Products-Radiation* ***(60)***.

Some polymers used in filter manufacturing have limited resistance to sterilizing irradiation. Filters may be constructed of radiation-resistant materials or polymers formulated with antioxidant additives designed to

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protect the polymer from excessive radiation-induced degradation. Irradiation sterilization, as with all sterilization methods, requires validation to establish sterility and stability.

Irradiation sterilization has several advantages, such as a high sterility-assurance level, no residual gas-or chemical-sterilization components, dry filters, and minimal interference by packaging components with the sterilization process. Disadvantages include potential lack of filter compatibility to high-dose and limited-filter shelf life (most irradiated filters are not validated to withstand a second sterilization cycle involving irradiation or steam). Steam sterilization of previously irradiated filters may increase extractables and leachables and could lead to loss of filter integrity, which could be confirmed by an integrity test.

## 8.3 Gas Sterilization

The most common sterilization gas is ethylene oxide, which may be used at 100% or diluted in a carrier gas. In addition to the increased environmental impact from using ethylene oxide as a sterilant, there may be safety factors such as the formation of ethylene chlorohydrin and ethylene glycol reaction products and ethylene oxide that may remain in the filter matrix. For successful filter sterilization by ethylene oxide, the filters should be dry and wrapped in a manner that allows for the free entry of water vapor and gas into the filter matrix. Following sterilization, the package also should permit the removal of residual gas and other undesirable byproducts. Due to the difficulty of ensuring gas penetration into and out of the filter membrane, this method of sterilization is typically not used for pleated filters but only for very small, encapsulated disc-style filters.

Parameters influencing gas sterilization are preconditioning factors, gas concentration, relative humidity, temperature, and time. Since there is no established correlation between lethality and the interrelationship between the sterilization parameters, destruction of a suitable biological indicator is the preferred method for validation and process monitoring. Spores of *Bacillus atrophaeus* have been established as the indicator of choice for this sterilization method.

## 8.4 Resterilization of Filters

Many sterilizing filters are designed to withstand multiple autoclave or in situ SIP cycles, either to demonstrate robustness or to meet filter-user resterilization requirements. Multiple gamma-irradiation sterilization cycles are generally not applicable due to cumulative dosage effects. Autoclave resterilization of previously radiation-sterilized filters may also not be possible. Suitability of resterilization processes should be confirmed with the filter manufacturer.

Where the resterilization of sterilizing filters is elected, the filter user should ensure that the resterilization process and the number of resterilization cycles does not adversely affect the filter or the product. Resterilization, including sterilization conditions, number of sterilization cycles and cumulative sterilization times should be validated.

Sterilizing filters should be routinely discarded after the processing of a single lot. In instances where repeated use can be justified, however, the sterilizing-filter validation including integrity-testing, bacterial retention studies, and cleaning should incorporate the maximum number of lots to be processed.

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# 10.0 Appendix I: Diffusive Flow Theory

The theory of diffusive flow requires a comprehensive discussion that may be found in reference literature ***(1-5)***; however, for the purposes of this report the following is provided.

When quantifying diffusive flow during integrity-testing of a wetted membrane, test gas movement (at sufficiently low pressures) follows well-established laws of diffusion. In a symmetric membrane, the diffusive flux of test gas to atmospheric pressure, as a function of the test pressure applied, may be described by:

**Equation 4**

$$N = \frac{DHP}{L}\varphi$$

**Where:**

$N$ = the diffusive flux of the test gas

$D$ = the diffusivity of the test gas through the wetting liquid

$H$ = the solubility coefficient of the test gas in the wetting liquid

$P$ = the applied differential pressure (or gauge pressure, if collected at atmospheric conditions)

$\varphi$ = the overall porosity of the structure

$L$ = the thickness of the wet layer (thickness of the membrane corrected by a "tortuosity" factor)

The diffusive flux, $N$, can be expressed in moles per unit area and unit time, but since these are measured at a fixed set of atmospheric pressure and temperature conditions, moles of gas can be converted to volumetric (ml/min or cc/min) units. Because the wetting fluid, test gas, filter thickness, porosity, and area are fixed, the expression for a volumetric diffusive flow in symmetric membranes further reduces to:

**Equation 5**

$$F = K_1 P$$

**Where:**

$F$ = the volumetric diffusive flow

$K_1$ = a proportionality constant

$P$ = the applied differential pressure (or gauge pressure, if collected at atmospheric conditions)

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Note that the diffusive flux of gas is independent of the actual filter-pore size, providing the pores are filled with the wetting liquid. Further, **Equation 5** predicts a linear relationship between the diffusive flow and the applied test pressure. This relationship ceases to exist if the applied test pressure exceeds that required to displace the wetting liquid with gas. Once the bubble-point pressure of the largest pore(s) is reached, bulk or viscous flow of air will occur in addition to the diffusive flow. This viscous flow of test gas through the pores from which the liquid has been displaced will obey Newton's laws of viscous transport, often modeled by the Hagen-Poiseuille equation for flow through cylindrical tubes.

**Equation 6**

$$Q = \frac{\pi P d^4}{128 \mu L}$$

**Where:**

$Q$ = the volumetric flow rate of the test gas

$P$ = the applied differential pressure (or gauge pressure, if collected at atmospheric conditions)

$d$ = the diameter of the pore

$\mu$ = the viscosity of the test gas

$L$ = the length test gas must travel to the downstream side, or the length of wetted pores through the membrane

The pressure at which a given pore will open to viscous flow can be estimated from the cylindrical capillary relationship attributed to Laplace, often referred to as the "Bubble Point equation."

**Equation 7**

$$P = \frac{4k\gamma cos\theta}{d}$$

**Where:**

$P$ = the differential pressure at which a given pore will open

$k$ = the correction factor for the shape of the largest pores

$\gamma$ = the surface tension of the wetting liquid

$cos\theta$ = the contact ("wetting") angle between the liquid and the membrane

$d$ = the diameter of the largest pores

To demonstrate the dependence on the liquid used to wet the pores and its interaction with the filter material, **Equation 7** shows an inverse relationship between the pore diameter and the test pressure necessary to free the wetting liquid from the pore. If the wetting liquid and membrane surface chemistry are held constant, the expression can be simplified to read:

**Equation 8**

$$d = \frac{K_2}{P}$$

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**Where:**

$d$ = the diameter of the largest pore

$K_2$ = a proportionality constant

$P$ = the differential pressure at which a given pore will open

$K_2$ is a correction factor accounting for shape of the largest pores as well as wetting properties for a given membrane/liquid combination. The value of the constant, and therefore the bubble point, in relationship to its retentive capabilities for a given contaminant, is established empirically.

The theory behind diffusive flow-testing can best be summarized by the extended integrity-test profile in **Figure 7.0-3**, depicting the gas-flow properties of a wetted filter as a function of the applied test pressure. The linear portion at the lower test pressures corresponds to the diffusive flow regime described by **Equation 4** or **Equation 5**, while viscous flow becomes the main transport mechanism for the steeper portion at higher pressures. The transition from diffusive to bulk flow (diffusive plus viscous flow) represents the maximum end of the pore-size distribution, as the larger pores are being voided of their wetting liquid. The relative size of the membrane's largest pores can be estimated from the test pressure using **Equation 8**.

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# 11.0 Appendix II: Nondestructive Physical Integrity Test Methods

## 11.1 Bubble Point Test

The bubble point test is based on the premise that membrane pores can be represented as a multiplicity of capillaries, and the knowledge that liquid is held in a capillary tube due to surface tension and other factors. The minimum gas pressure required to force liquid out of the pores is a direct function of pore diameter. The bubble point test is used to determine the pressure at which there is bulk flow of the test gas through the largest pores in the membrane that are the first from which liquid is displaced. Since the observed bubble point is a function of the wetting-liquid surface tension, the contact angle, the pore shape, and the pore size, results are specific to each membrane polymer/surface chemistry, structure, wetting fluid combination, methods, equipment and surface area. While bubble point is an indication of the largest pores, it cannot be used directly to calculate their diameter. The bubble point test also does not indicate the number of largest pores, which can affect the probability of bacterial retention (if the organisms are smaller than the pores).

The test can be performed manually by subjecting a wetted filter membrane to gas pressure on the upstream side. Tubing on the membrane's downstream side is immersed in a liquid. The differential pressure at which a steady stream of bubbles, or in some cases the displacement of liquid, is observed is the bubble point. The typical test equipment for performing a manual bubble point test is illustrated in **Figure 11.0-1**.

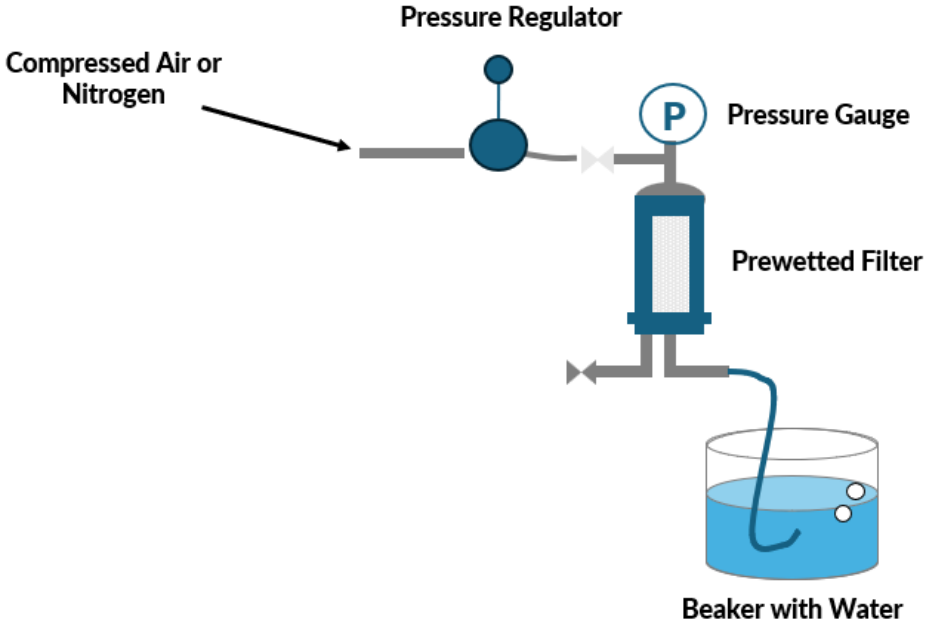


**Figure 11.0-1 Typical Manual Bubble Point Test Equipment**

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The manual bubble point test is a sensitive, visual technique for disc filters and other small surface-area membrane filters; it may be performed as one of the quality assurance tests for membranes during the casting process. The bubble point test detects major and minor defects and out-of-size pores and can be correlated with the bacterial retention study. It is also a reliable integrity test of larger surface-area filters until the diffusional flow across the membrane masks the flow rate from the pores from which the wetting fluid has been displaced.

The following are points to consider when performing the bubble point test manually:

- Use a pressure gauge with easy-to-read incremental steps and increase slowly in a step-wise manner.
- Allow pressure to stabilize at each step in the pressure increase. The true bubble point value may be overshot if pressure is increased too rapidly. The test cannot be repeated without rewetting the membrane.
- Minimize downstream connections and avoid kinking the downstream tubing.
- Check for leaks in the system.
- Clearly define the bubble point as bulk air flow. The most common error is misidentifying bubbles that are not the result of bulk flow. Some bubbles will appear because of diffusion across the wetted membrane. Free flowing air is the indication, not the first bubble.
- Manually performed, bubble point-test determination can be subjectively different from one operator to another; therefore, operators should be properly trained to conduct the test and to interpret test results.
- Keep upstream volume to a minimum. With extremely high upstream volumes, longer stabilization between pressure increases may be required.
- Maintain the temperature within the specified range.

Test attributes:

- Relatively fast, with a short stabilization time
- Can be directly related to the membrane pore size
- Relatively easy to perform for small-to-medium area filters
- Easy to establish correlation between bacterial retention and physical integrity test values
- Sensitivity may be limited for large-membrane-area systems if diffusive air flow below the bubble point is significant
- Manual testing requires downstream (i.e., sterile side) manipulations

## 11.2 Diffusive Flow Test

In a wetted membrane filter under pressure, gas molecules migrate through the liquid-filled pores at differential pressures below the bubble point via a diffusion process following Fick's Law of Diffusion. The overall rate of diffusion for a filter is proportional to the applied test pressure, its membrane surface area and wetted thickness, and the solubility and diffusivity of the test gas in the wetting liquid. In small surface-area filters, such as flat-disc filters, this gas flow may be too low to be measured accurately. In large surface-area filter systems, it is significant and can be measured to perform an integrity test. To accurately measure the test limit, one must use the correct limit based on the gas type. This only applies to Diffusion/Pressure Hold test; the bubble point test is not influenced by gas type.

The diffusive flow test quantitatively measures the sum of diffusive flow through the membrane and flow through any open pores. The filter user should use the manufacturer's recommended testing method for each filter, ensuring that the method is appropriate and has been validated for the specific case. Diffusive flow

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measurements may be performed downstream of the wetted membrane under constant test pressure, or on the upstream side by measuring the air flow required to maintain constant test pressure. These tests are performed on membrane filters by first wetting out the filter with an appropriate liquid and draining excess fluid. The upstream side of the filter is pressurized, and the diffusive flow is measured. Test equipment for a diffusive-flow integrity test with flow measurement on the downstream side is illustrated in **Figure 11.0-2**. The inverted burette or measuring cylinder filled with liquid for the collection of diffused gas can be replaced with an appropriate gas flowmeter.

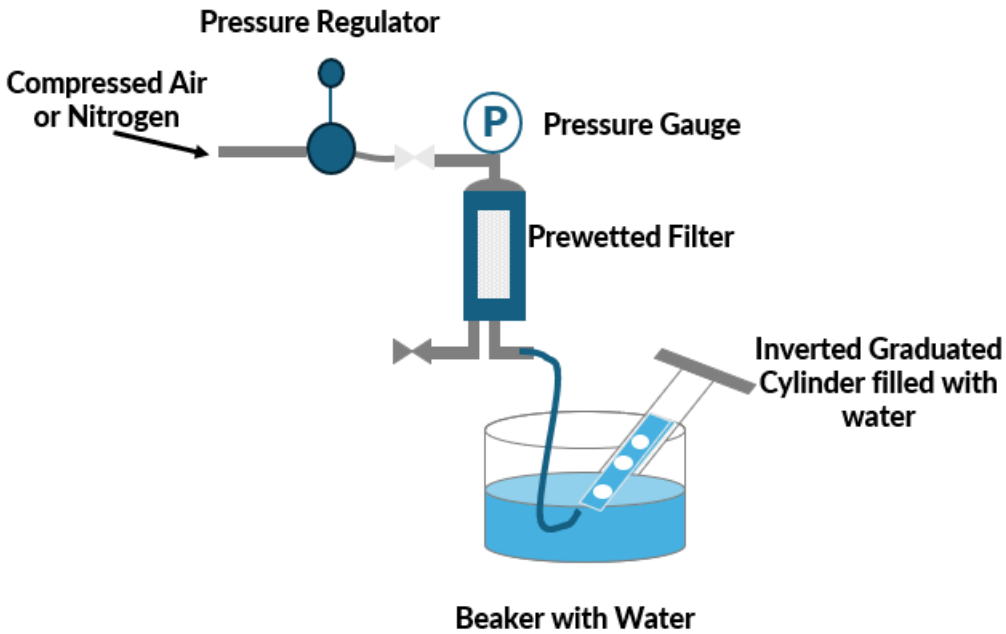


**Figure 11.0-2 Typical Test Setup for Manually Determined Diffusive Flow Rates**

The following are points to consider when performing the diffusive flow test manually:

- Minimize downstream connections and downstream volume.
- Avoid kinking downstream tubing.
- Check for leaks in the system.
- Use a volume-collection vessel with the proper resolution.
- Use downstream tubing with the smallest practical diameter and shortest practical length to ensure proper flow resolution.
- Keep upstream volume to a minimum; with extremely high upstream volumes, proper stabilization time must be allowed.
- Maintain a stable testing temperature throughout the testing period.
- Allow sufficient equilibration time after pressurizing the system.
- The test should be performed at the filter manufacturer's specified pressure.

Test attributes:

- Developed to provide increased sensitivity for large membrane surface-area systems
- Provides an objective, quantitative measure of gas flow
- Can be correlated to bacterial retention (as discussed in **Section 7.1**)
- May be sensitive to temperature fluctuations
- Manual testing requires downstream (i.e., sterile side) manipulations

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## 11.3 Pressure Hold/Decay Test

The pressure hold (or pressure decay) test is an indirect method of upstream diffusive flow-testing. In this method, the filter-housing is pressurized to a predetermined setting and then isolated from the pressure source. Gas flow across the membrane is quantitatively measured as a decay in pressure over a specific period of time. This test has been indirectly correlated to bacterial retention through its relationship to the diffusive flow test.

The allowable pressure drop is not published by filter manufacturers because it is unique to a given filter assembly and piping system, and it is calculated by taking into account the diffusive-flow specification at a given pressure and constant temperature, the upstream system volume, and the duration of the test. Therefore, this test requires a determination of the upstream volume of the system and a calculation of the allowable pressure drop.

The pressure hold/decay test measures the diffusive gas (typically air or nitrogen) flow through all the wetted pores, and the bulk gas flow through larger non-wetted pores, when applying a predetermined gas pressure to a wetted filter in a system that has no upstream leaks. Pressure decay on the upstream side is then measured, after isolating the incoming gas supply. Upstream integrity tests are particularly useful in critical fluid applications since they can be performed without compromising the sterility of the downstream system.

The pressure decay is a function of the gas flow through the filter and the upstream volume of the filter system. The pressure hold/decay test will confirm the integrity of the entire assembly, including housing seals. Therefore, failure analysis for lack of seal integrity is required prior to reaching the conclusion that the filter element itself is not integral.

If the pressure decay is below the maximum allowable value, the filter passes the physical integrity test. If the decay and concomitant pressure drop exceed the correlated maximum-allowable value, the filter fails the test, and an investigation should be performed (see **Section 7.6**).

In performing the pressure hold/decay test, the physical conditions must be held constant, as the test is affected by interactions of temperature, system volume and filter area. The pressure hold/decay test equipment is illustrated in **Figure 11.0-3**. This test commonly is performed preprocess, post-sterilization of the filter assembly, and again after sterilizing filtration.

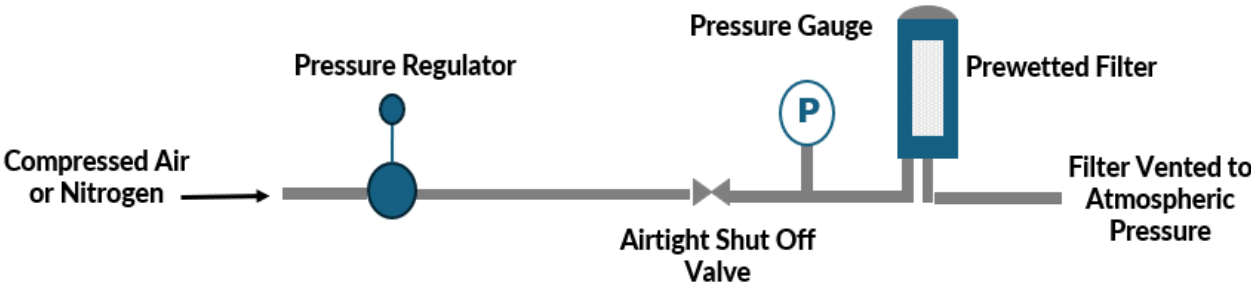


**Figure 11.0-3 Typical Test Setup for the Pressure Hold/Decay Test**

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The following are points to consider when performing the pressure hold/decay test:

- There should be no leaks in the upstream side of the system.
- Upstream volume should be determined to ensure a proper maximum-allowable-pressure drop calculation. It is recommended that upstream volume be kept to a minimum.
- The downstream side of the filter should be vented to the atmosphere.
- Sufficient stabilization time should be allowed before beginning the test; stabilization time increases with larger upstream volumes and increased filter surface-areas.
- A pressure gauge or transducer with the proper degree of accuracy for determining the pressure drop during testing should be used.
- Test time may need to be extended if the pressure-drop rate is extremely small relative to the sensitivity of the gauge or transducer.
- The specified temperature should be maintained to ensure that the temperature does not change during the integrity test. Note that lower upstream volumes will result in increased sensitivity to temperature changes.

Test attributes:

- Relatively easy to perform
- No downstream manipulation is required
- Can be correlated to bacterial retention (as discussed in **Section 7.1**)
- Sensitivity of the test decreases as the upstream volume of the system is increased
- Resolution of the test is limited by the resolution of the pressure sensor used
- Process conditions for the test, including temperature and gas used for pressurization, should be defined for the issuance of integrity test values.

## 11.4 Automated Integrity-Test Instruments

Numerous automated integrity-test instruments are available for use in the pharmaceutical industry to determine a bubble-point pressure value using incremental pressure hold/decay tests. The pressure hold/decay tests are performed at increasing pressures, with the bubble point identified where the pressure decay become nonlinear, based on an algorithm specific to each instrument manufacturer. Automated integrity-test units determine diffusive-flow values, either by measuring pressure hold/decay at a specified test pressure and calculating the diffusive-flow value using a predetermined measurement of the upstream volume of the filter assembly or by direct-flow measurement using an upstream mass flow meter. Another method to determine diffusive-flow values is a system with additional small volumes to maintain constant test pressure during the test, which eliminates the need to measure upstream volume.

Irrespective of the technique used to determine filter bubble point or diffusive flow, the automated integrity-test instrument must be qualified against an independent reference method.

## 11.5 Correlation Between Pressure Hold/Decay and Diffusive Flow

Diffusive flow prior to the initiation of any bulk flow through the first opened pore can be expressed by **Equation 10**, derived from Fick's Law and Henry's Law.

**Equation 10**

$$V_S = \left(\frac{M}{\rho_S}\right) A_F \frac{D\,(p_2 - p_1)k}{H_c L}$$

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The measured flow resulting from a specified differential pressure can be related to the microbial retentive capability of a filter membrane.

The relationship between this flow, measured by the nondestructive pressure hold/decay test and the destructive microbial challenge test, is the objective of the validation for each filter medium and each filter configuration.

**Equation 10** shows that diffusive flow is proportional to the difference between the test pressure and atmospheric pressure. Since the pressure in the upstream volume decreases, due to the effluent diffusive flow, the flow also decreases.

From the ideal gas equation,

**Equation 11**

$$p_2 V_{up} = NRT$$

the change of state per time, *dt,* can be derived:

**Equation 11a**

$$\frac{dp_2}{dt} V_{up} = \frac{dN}{dt} RT$$

With

**Equation 12**

$$-V_S \rho_S = \frac{dN}{dt}$$

and **Equation 10** and **Equation 11a** can be written as:

**Equation 13**

$$\frac{-dp_2}{dt} \frac{V_{up}}{RT} = \frac{MA_F D(p_2 - p_1)k}{LH_c}$$

Hence,

**Equation 13a**

$$\frac{1}{p_2 - p_1}(-dp_2) = \frac{RT}{V_{up}} MA_F \frac{D}{LH_c} kdt$$

And carrying out the integration between limits and simplifying yields the result:

**Equation 13b**

$$p_{2I} - p_{2II} = (p_{2I} - p_I)\left(1 - exp\left[\frac{-RT}{V_{up}} MA_F \frac{D}{LH_c} k(t_{II} - t_I)\right]\right)$$

To obtain an equation for the pressure decay (as a function of volume flow in the standard state), introducing **Equation 10** in **Equation 13b** and using $pS = \rho_S RT/M$ yields:

**Equation 14**

$$p_{2I} - p_{2II} = (p_{2I} - p_I)\left(1 - exp\left[\frac{-(t_{II} - t_I)}{V_{up}} \frac{V_S \rho_S}{p_{2I} - p_I}\right]\right)$$

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Measurements of pressure and time are necessary for the determination of $V_s$ because the retentive capability of the filter membrane is related to the flow $V_s$, as shown in **Equation 14**. Additionally, the upstream volume must be known and the temperature in the upstream volume must be constant.

**Table 11.5-1 Nomenclature for Symbols Used in Equations 10-14**

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $A_F$ | area of the filter | $p_{2I}$ | pressure at $t_I$ |
| D | diffusion constant for gas/liquid system | $p_{2II}$ | pressure at $t_{II}$ |
| $H_c$ | Henry's Law coefficient | $p_{2I} - p_{2II}$ | pressure decay |
| k | membrane specific correction factor | $p_{2I} - p_I$ | entered test pressure |
| L | length of diffusion path | R | ideal gas constant |
| N | moles | T | temperature |
| M | molar mass | $t_{II} - t_I$ | entered test time |
| $p_1$ | atmospheric pressure | $V_{up}$ | upstream volume |
| $p_2$ | upstream pressure | $V_s$ | standard volume flow |
| $p_2 - p_1$ | differential pressure | $\rho_S$ | density of gas in the standard state |

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# 12.0 Appendix III: Integrity Test Troubleshooting Guide

Problems may occur when performing integrity tests, manually or with an automatic test unit. The following tables are a troubleshooting guide to facilitate failure investigations. It is important to note that an automatic test device avoids human subjectivity and, therefore, may be more reproducible.

When investigating a manual integrity test failure, the following items should be considered:

- Insufficiently wetted filter
- Inappropriate wetting medium (e.g., solvent instead of water)
- Incorrect filter pore size rating

**Table 12.0-1** lists items for consideration when investigating an automatic device integrity-test failure.

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**Table 12.0-1 Automatic Device Integrity Test Troubleshooting Guide**

| Issue | Possible Cause |
|---|---|
| **No Test Pressure Build-Up** | • Filter system leakage (e.g., damaged sealing, damaged tubing, valve open, clamps improperly closed, damaged filter)<br>• Improperly wetted filter<br>• Inappropriate wetting medium (e.g., solvent instead of water)<br>• Filter contains traces of product solution or extraneous material from wetting fluid<br>• Incorrect filter pore-size rating<br>• Gas supply to the unit blocked<br>• Inlet tubing to the filter-housing improperly installed<br>• Excessive temperature drifts<br>• Improper valve function within the test unit<br>• Internal pneumatic leaks |
| **Continuous Venting of the Unit** | • Internal valve contamination and improper function |
| **Abortion of Test During Stabilization Phase** | • Filter system leakage (e.g., damaged sealing, damaged tubing, valve open, clamps improperly closed, damaged filter)<br>• Insufficiently wetted filter<br>• Contaminated filter<br>• Incorrect filter pore-size rating<br>• Excessive temperature drifts<br>• Internal pneumatic leaks |
| **Abortion During Upstream Volume Test** | • Upstream volume of filter system is too large<br>• External reference volume is connected inappropriately<br>• Internal pneumatic leaks<br>• Filter failure<br>• Inadequate stabilization time<br>• Insufficiently wetted filter |
| **Measured Volume is Wrong** | • Incorrect input of reference volume<br>• Contaminated internal refence tank<br>• Internal pneumatic leaks |

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| Issue | Possible Cause |
|---|---|
| **Failure of Bubble Point** | • Damaged filter<br>• Insufficiently wetted filter<br>• Inappropriate wetting medium (e.g., solvent instead of water)<br>• Wetting fluid temperature<br>• Incorrect filter pore-size<br>• Filter contains traces of product solution or extraneous material from wetting fluid<br>• Internal pneumatic works improperly<br>• Inlet tubing to the filter-housing improperly installed<br>• Incorrect test gas<br>• Incorrect test parameter setting<br>• Incorrect test code<br>• Large multiple-cartridge assembly |
| **Failure of Diffusive Flow** | • Damaged filter<br>• Insufficiently wetted filter<br>• Inappropriate wetting medium (e.g., solvent instead of water)<br>• Filter contains traces of product solution or extraneous material from wetting fluid<br>• Incorrect filter pore size<br>• Internal pneumatic works improperly<br>• Inlet tubing to the filter housing improperly installed<br>• Incorrect test parameter setting<br>• Incorrect test gas<br>• Temperature shifts during test time<br>• Insufficient stabilization time |

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| Issue | Possible Cause |
|---|---|
| **Failure of Pressure Decay** | • Filter-system leakage (e.g., damaged sealing, valve open, clamps improperly closed, damaged filter)<br>• Insufficiently wetted filter<br>• Inappropriate wetting medium (e.g., solvent instead of water)<br>• Incorrect filter pore-size rating<br>• Filter contains traces of product solution or extraneous material from wetting fluid<br>• Internal pneumatic works improperly<br>• Inlet tubing to the filter-housing improperly installed<br>• Incorrect test-parameter setting<br>• Incorrect test gas<br>• Temperature shifts during test time<br>• Insufficient stabilization time or incorrect test time |
| **No Pressure Decay** | • Inlet tubing to the housing improperly installed<br>• Downstream valve closed<br>• Blocked connector<br>• Blocked internal pneumatic<br>• Elevated temperature during test time |
| **Failed or Elevated Diffusion Test Result** | • Damaged filter<br>• Insufficiently wetted filter<br>• Inappropriate wetting medium (e.g., solvent instead of water)<br>• Incorrect filter pore size<br>• Internal pneumatic works improperly<br>• Inlet tubing to the filter-housing improperly installed<br>• Incorrect test-parameter setting<br>• Incorrect test gas<br>• Temperature shifts during the test time<br>• Insufficient stabilization time<br>• Incorrect upstream volume (i.e., when not automatically accounted for) |

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# Technical Report No. 26: Sterilizing Filtration of Liquids

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